Society for Computer Technology & Research 's (SCTR's)

Pune Institute of Computer Technology (PICT), Pune

An Autonomous Institute affiliated to the Savitribai Phule Pune University (SPPU)
Approved by AICTE & Government of Maharashtra,
Accredited by NAAC (A+) & NBA [All eligible UG Programs]

Syllabus for

S.Y. B. Tech Computer Engineering

(2026-27 Course)

With effect from (AY 2026-27)
National Education Policy (NEP) 2020 Compliant
Approved by the Board of Studies (BoS) and Academic Council

Abbreviations used

Sr. No.	Broad Category of the course	Sub- Category of course	Category Code
I.	**Basic Science/ Engineering Science Course (BSC/ ESC)**	Basic Science Course (BSC)	**BS**
		Engineering Science Course (ESC)	**ES**
II.	**Program Courses (PC)**	Program Core Course (PCC)	**PC**
		Program Elective Course (PEC)	**PE**
III.	**Multidisciplinary Courses (MC)**	Multidisciplinary Minor (MDM)	**MD**
		Open Elective (OE)	**OE**
IV.	**Skill Courses (SC)**	Vocational and Skill Enhancement Course (VSEC)	**VS**
V.	**Humanities Social Science and Management (HSSM)**	Ability Enhancement Course (AEC-01, AEC-02)	**AE**
		Entrepreneurship/Economics/ Management Courses (EEM)	**EE**
		Indian Knowledge System (IKS)	**IK**
		Value Education Course (VEC)	**VE**
VI.	**Experiential Learning Courses (ELC)**	Research Methodology (RM)	**RM**
		Community Engagement Project (CEP) / Field Project (FP)	**CE**
		Project (PRJ)	**PR**
		Internship/ On Job Training (IP/OJT)	**IP**
VII.	**Liberal Learning Courses (LLC)**	Co-curricular Activities (CCA)	**CC**
VIII.	**Honors Courses (HC)**	offered within the same branch depth in the same branch (advanced specialization & research).	**HC**
IX.	**Minor Courses (MIC)**	offered by other branches **breadth across disciplines (interdisciplinary exposure, applied projects).**	**MC**
X.	**Double Minor Courses (DMC)**	Additional minor to be offered by other branches **breadth across disciplines (interdisciplinary exposure, applied projects).**	**DM**

Detailed guidelines for examination:

Link: Guidelines for examination

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Semester - III

Semester 3			Teaching Scheme (Hours/Week)				Credits/ Grades				Examination Scheme and Marks						
Broad Category of Course	Subject code	Name of subjects	L	P	T	Total	L	P	T	Total	Theory			Practical			Total
											CIE	ISE	ESE	CIE	ESE		
											[20]	[20]	[60]	TW	(PR)	(OR)	
PCC	2613PC11	Data Structures (DS)	3	-	-	**3**	3	-	-	**3**	20	20	60	-	-	-	**100**
PCC	2613PC12	Computer Organization and Architecture (COA)	3	-	-	**3**	3	-	-	**3**	20	20	60	-	-	-	**100**
PCC	2613PC13	Discrete Mathematics (DM)	3	-	-	**3**	3	-	-	**3**	20	20	60	-	-	-	**100**
PCC	2613PC31	Data Structures Lab (DSL)	-	4	-	**4**	–	2	-	**2**	-	-	-	50	50	-	**100**
PCC	2613PC32	Computer Organization and Architecture Lab (COAL)	-	2	-	**2**	-	1	-	**1**	-	-	-	25	-	50	**75**
MDM	2603MD11	MDM - 1	2	-	-	**2**	2	-	-	**2**	20	20	60	-	-	-	**100**
MDM	2603MD51	MDM - 1 #	-	-	1	**1**	-	-	1	**1**	-	-	-	25	-	-	**25**
EEM	2613EE11	Engineering Economics and Financial Management (EEFM)	1	-	-	**1**	1	-	-	**1**	-	-	-	25	-	-	**25**
AEC	2603AE31	Professional Development and Career Readiness (PDCR)	-	2	-	**2**	-	1	-	**1**	-	-	-	25	-	-	**25**
OE	2603OE5X	Foreign Language Studies (FLS)	-	-	2	**2**	-	-	2	**2**	-	-	-	50	-	-	**50**
VEC	2603VE11	Universal Human Values (UHV)	2	-	-	**2**	2	-	-	**2**	-	-	-	25	-	-	**25**
CEP	2603CE31	Community Engagement Project (CEP)	-	2	-	**2**	-	1	-	**1**	-	-	-	25	-	-	**25**
Total			**14**	**10**	**3**	**27**	**14**	**5**	**3**	**22**	**80**	**80**	**240**	**250**	**50**	**50**	**750**

L: Lecture, **P:** Practical, **T:** Tutorial,
ISE: In-Semester Examination, **CIE:** Continuous Internal Evaluation, **ESE**: End-Semester Examination,
TW:Term work, **PR:** Practical Examination, **OR**: Oral Examination .

Semester- IV

Semester 4			Teaching Scheme (Hours/Week)				Credits/ Grades				Examination Scheme and Marks						
Broad Category of Course	Subject code	Name of subjects	L	P	T	Total	L	P	T	Total	Theory			Practical			
											CIE	ISE	ESE	CIE	ESE		Total
											[20]	[20]	[60]	TW	(PR)	(OR)	
PCC	2614PC11	Software Engineering (SE)	3	-	-	**3**	3	-	-	**3**	20	20	60	-	-	-	**100**
PCC	2614PC12	Database Management Systems (DMS)	3	-	-	**3**	3	-	-	**3**	20	20	60	-	-	-	**100**
PCC	2614PC13	Operating Systems (OS)	3	-	-	**3**	3	-	-	**3**	20	20	60	-	-	-	**100**
PCC	2614PC31	Operating Systems Lab (OSL)	-	2	-	**2**	-	1	-	**1**	-	-	-	25	-	25	**50**
PCC	2614PC32	Database Management Systems Lab (DMSL)	-	4	-	**4**	-	2	-	**2**	-	-	-	50	25	-	**75**
MDM	2604MD11	MDM - 2	2	-	-	**2**	2	-	-	**2**	20	20	60	-	-	-	**100**
MDM	2604MD31	MDM - 2 #	-	2	-	**2**	-	1	-	**1**	-	-	-	25	-	-	**25**
OE	2604OE51	Open Elective - II *	-	-	2	**2***	-	-	2	**2**	-	-	50	-	-	-	**50**
VSEC	2614VS31	Project Based Learning (PBL)	-	4	-	**4**	-	2	-	**2**	-	-	-	50	-	25	**75**
EEM	2614EE11	Entrepreneurship (EP)	1	-	-	**1**	1	-	-	**1**	-	-	-	25	-	-	**25**
AEC	2604AE31	Collaborative Skills, Digital Ethics, and Cyber Security (CDC)	-	2	-	**2**	-	1	-	**1**	-	-	-	25	-	-	**25**
VEC	2604VE11	Indian Constitution and Social Responsibility (ICSR)	1	-	-	**1**	1	-	-	**1**	-	-	-	25	-	-	**25**
Total			**13**	**14**	**2**	**29**	**13**	**7**	**2**	**22**	**80**	**80**	**290**	**225**	**25**	**50**	**750**

#: Tutorial or laboratory as applicable. Choose one course from MDM Baskets. MDM:X is basket number;. **Refer Annexure-1** for MDM details.

***:** Open Elective (OE) offered by online platforms such as SWAYAM/ NPTEL, **Refer Annexure-II** for details.

X: Serial number of the courses under that particular category.

Second Year B-Tech
(S. Y. B. Tech)
Semester-3

Second Year B. Tech (S. Y B. Tech) AY (2026-27)
Computer Engineering (CE)
[2613PC11]: Data Structures (DS)

Semester	Credits	Teaching Scheme	Examination Scheme
3	3	L: 3 Hrs./ Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- C Programming for Problem Solving (CPPS)
- Object Oriented Programming Using C++/Java (OOPC).

Course Objectives: The objective of this course is to provide students with

- Basic techniques of algorithm performance analysis.
- Knowledge of various data searching and sorting methods with pros and cons.
- Structural constraints and advantages in usage of the data using Hashing.
- Data manipulation using various data structures like Linked List, Stack, Queues.
- Data handling with tree data Structures.

Course Outcomes: After completing this course, students will be able to:

CO1: Evaluates and compares the time and space complexities of various searching and sorting algorithms and Hashing techniques using Asymptotic notation.

CO2: Write pseudocode for fundamental operations like insertion, deletion, traversal, and searching using static (Array) and dynamic (Linked List) structures to store and process the structured data.

CO3: Solve the given infix arithmetic expression in well-parenthesized form by converting it into postfix expression using primitive operations of Stack and solve job scheduling problems using Queue to show simulation of job scheduling in CPU.

CO4: Write pseudocode for recursive and non-recursive functions for fundamental operations like insertion, deletion, traversal and searching data in binary search tree, height balanced tree.

COURSE CONTENTS

Module-I	Introduction to Analysis of Algorithms	12 Hrs.

Introduction of Algorithms, Analysis of Algorithms, Complexity of algorithms- Space complexity, Time complexity.

Data Structures: Abstract Data Types (ADT), Concept of linear and Non-linear data structures. Use of Array and Associative/Jagged Arrays.

Searching: Search Techniques, Sequential search, Sentinel search, Binary search, Fibonacci search.

Sorting: Types of Sorting- Internal and External Sorting, Stable Sorting, Sorting methods- Bubble sort, Insertion sort, Selection sort, Quick sort, Merge sort, Comparison of All Sorting Methods.

Hashing: Types of Hashing, Hash table, Hash functions, Collision resolution strategies.

Module-II	Linked List	9 Hrs.

Static memory allocation vs Dynamic Memory Management, linked list using dynamic memory management, **Linked List-** Abstract Data Type, linked list operations, Types of linked list- Linear, Circular linked lists and Doubly Linked List and operations, doubly circular linked list, Polynomial representation using Linked List, Generalized Linked List (GLL) concept and use of GLL for multi-variable polynomial

Module-III	**Stacks & Queues**	**9 Hrs.**

Stack an Abstract Data Type, Representation of Stacks Using Sequential Organization, stack operations, Multiple Stacks, Applications of Stack-Polish notation, need for prefix and Postfix expressions, Expression Evaluation and Conversion, Postfix expression evaluation, Linked Stack and Operations.

Recursion- concept, variants of recursion- direct, indirect, head, tail.

Queue as Abstract Data Type, Realization of Queues Using Arrays, Circular Queue, Advantages of circular queues. Multi-queues, Deque, Priority Queue, Linked Queue and operations. Applications of Queue- Job Scheduling.

Module-IV	**Trees**	**9 Hrs.**

Definition & Basic Terminology of Tree, Binary Tree, Full Binary Tree, Complete Binary Tree, Binary Search Tree and primitive operations, Threaded Binary Tree, Heap Tree, AVL tree and its operations. Introduction to B Tree & B+ Tree.

Text Books:

T1.	Horowitz and Sahani, *Fundamentals of Data Structures in C++*, University Press, 2007, 2nd Edition, ISBN- 978-07-1678-292-6.
T2.	Goodrich, Tamassia, Goldwasser, *Data Structures and Algorithms in Python*, Wiley Publication 2021,1st Edition, ISBN- 978-93-5424-786-6.
T3.	Herbert Schildt, *Java: The Complete Reference*, McGraw Hill Education, 2014, 9th Edition, ISBN- 978-0749467241.

Reference Books:

R1.	Brassard & Bratley, *Fundamentals of Algorithmics*, Prentice Hall India/Pearson Education, 1996, 2nd Edition, ISBN- 978-8120311312.
R2.	R. Gilberg, B. Forouzan, *Data Structures: A Pseudocode approach with C++*, Cengage Learn, 2005, 2nd Edition, ISBN- 978-8131503140.
R3.	M. Weiss, *Data Structures and Algorithm Analysis in C++*, 2nd edition, Pearson Education, 2002, ISBN- 978-0201498400.

Relevant MOOCs Course (Course name and Weblink)

1. NPTEL course: Programming, Data Structures And Algorithms Using Python, By Prof. Madhavan Mukund, Chennai Mathematical Institute, Link: https://onlinecourses.nptel.ac.in/noc26_cs79/preview
2. NPTEL course: Design and analysis of algorithms, By Prof. Madhavan Mukund, Chennai Mathematical Institute, Link: https://onlinecourses.nptel.ac.in/noc26_cs67/preview
3. IITBombayX: Foundations of Data Structures: https://www.edx.org/learn/data-structures/iitbombay-foundations-of-data-structures

Relevant Topics for Self-study:

Interpolation Search, Bucket Sort, Shell Sort, Heap Sort, Applications of Linked List, Applications of balanced search trees.

Second Year B. Tech (S. Y B. Tech) AY (2026-27)
Computer Engineering (CE)
[2613PC12]: Computer Organization and Architecture

Semester	Credits	Teaching Scheme	Examination Scheme
3	3	L: 3 Hrs./ Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of
- Digital Electronics and Logic Design

Course Objectives: The objective of this course is to provide students with
- Basics of structure, function , characteristics of computer Organization.
- Instruction set and logic to build Assembly Language Programs (ALP).
- Knowledge of instruction level parallelism, parallel organization of multiprocessor,multithreading .
- Multiple processor organizations, memory architectures (UMA, NUMA, CC-NUMA) and multicore systems.

Course Outcomes: After completing this course, students will be able to:

CO1: Describe the computer architecture, basic components (CPU, Main Memory, I/O Module, Registers), phases of instruction execution and floating point representation.

CO2: Use efficient addressing mode which result functionally correct code for performing arithmetic and logical operations using different data types.

CO3: Illustrate concept and design principles of memory systems-cache, virtual and main memory.

CO4: Distinguish multiple processor organizations (SSID, SIMD, MISD, MIMD) and memory architectures.

CO5: Analyze different approaches to multithreading and explore multicore systems.

COURSE CONTENTS

Module-I	Introduction to Processor Architecture	12 Hrs.

Basic structures of computers-Functional units, performance Measurement. Evolution of computer architecture. Processor Structure and Function - Processor Organization, Register Organization, The Instruction cycle, instruction pipeline.

Computer Arithmetic: Integer and Floating-point representation and arithmetic.

Instruction sets: Characteristics and function, Data types, addressing modes and formats, assembler directives, execution of program. Instruction types-data transfer, arithmetic, logical, branch, call, ret.

Internal Memory Technology: DRAM operation, row decoder column decoder chip organization, error correction code (single and double).

Reduced instruction set computer: Instruction execution characteristics, RISC architecture, RISC pipeline, RISC Vs CISC.

Module-II	Memory organization and management	9 Hrs.

Memory system overview: characteristics of memory system, memory hierarchy. Cache Memory – Basics, cache memory principles, elements of cache design, cache coherence and MESI protocol.

Virtual Memory: main memory allocation, Virtual to Physical Address Translation, Paging, Page Faults, TLB.

Case study: Memory management in advanced processors.

Module-III	Parallel processing and Multicore architecture	9 Hrs.

Multiple Processor Organization: SISD, SIMD, MISD, MIMD. Uniform memory access (UMA), Non uniform memory access (NUMA), CC-NUMA. Multicore: Hardware and software performance issues, need of multicore. Multicore organization, heterogeneous multicore organization: CPU and GPU.

Case study: Intel Core i7 5960X

Module-IV	Multicore programming	9 Hrs.

Software developer viewpoint for multicore programming. Challenges of multicore programming.
Multithreading: Thread, comparing thread to processes, Approaches of Multithreading.
Communication and synchronization of concurrent tasks: Communication and synchronization, synchronizing concurrency, thread Strategy Approaches

Case study: POSIX 1. Write an application to demonstrate inter process communication.
2. Write a multithreading application for calculators.

Text Books:

T1.	W. Stallings, *Computer Organization and Architecture: Designing for Performance*, Pearson, 2019, 11th Edition, ISBN- 10:0-13-499719-0.
T2.	A. Ray, K. Bhurchandi, *Advanced Microprocessors and peripherals: Arch, Programming, Interfacing*, Tata McGraw Hill, 2016, 3rd edition, ISBN- 978-1259006135.
T3.	James Turley, *Advanced 80386 Programming Techniques, McGraw-Hill*, 2015, 3rd edition, ISBN- 978-0078813429.
T4.	*Intel® 64 and IA-32 Architectures Software Developer's Manual (Vol. 1-3)* - Intel SDM, Intel VT-x & SGX Whitepapers - Virtualization & security documentation.

Reference Books:

R1.	C. Hamacher, V. Zvonko, S. Zaky, *Computer Organization*, McGraw Hill, 2002, 5th edition, ISBN- 978-1259005275.
R2.	Cameron Hughes, Tracey Hughes, *Multicore Programming Design and Implementation for C++ Developers* , Wiley, 2008, 1st Edition, ISBN- 978-0-470-28962-4.
R3.	Ray Seyfarth, *Introduction to 64-bit Intel Assembly Language Programming for Linux*, Createspace Independent Publishing Platform, 2014, 2nd Edition, ISBN- 9781478119203.
R4.	https://www.intel.com/content/dam/www/public/us/en/documents/datasheets/core-i7-900-ee-and-desktop-processor-series-datasheet-vol-1.pdf,

Relevant MOOCs Course (Course name and Weblink)

1. NPTEL course on, Microprocessor, By Dr. Pramod Agarwal, IIT Roorkee, Link: https://nptel.ac.in/courses/108107029/
2. NPTEL course on C programming and assembly language, By Prof. Janakiraman, IIT Madras, Link: https://onlinecourses.nptel.ac.in/noc23_cs93/preview
3. NPTEL course on, Computer Architecture, By Dr. Indranil Sengupt, Prof. Kamalika Datta, IIT Kharagpur, Link: https://nptel.ac.in/courses/106105163

Relevant Topics for Self-study:

1. Number System and their conversions.
2. System calls.
3. Handling Procedure and Macros using 64-bit Assembly Language Programming.

Second Year B. Tech (S. Y B. Tech) AY (2026-27)
Computer Engineering (CE)

[2613PC13]: Discrete Mathematics (DM)

Semester	Credits	Teaching Scheme	Examination Scheme
3	3	L: 3 Hrs./ Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Linear Algebra, Basic Mathematics.

Course Objectives: The objective of this course is to provide students with

- Fundamentals of set theory and counting principle.
- Concepts of relations and functions.
- Role of mathematics in problem solving .
- Concept of algebraic systems.

Course Outcomes: After completing this course, students will be able to:

CO1. Select suitable proof techniques to explain the reasoning clearly for a set of propositions by following standard structure.
CO2. Determine the correct type of relation and function based on properties and perform operations for a given application.
CO3. Solve the problems- Traveling Salesperson, Minimum Spanning Tree, Shortest Path and Huffman coding by selecting suitable algorithms in graph theory with correct methodology and accuracy.
CO4. Evaluate the algebraic structures into Semigroup, Group, Rings using properties of binary operations.

COURSE CONTENTS

Module-I	Set Theory and Counting	12 Hrs.

Set Theory: Introduction to Sets– Set Operations, Cardinality of set, Principle of inclusion and exclusion, Types of Sets, Power set, Methods of Proof-Proof by Contradiction, Proof by Mathematical Induction.

Propositional Logic- logic, Propositional Equivalences, Application of Propositional Logic- Translating English Sentences.

Counting: The Basics of Counting, rule of Sum and Product, Permutations and Combinations.

Module-II	Relations and Functions	9 Hrs.

Relations: Relations and their Properties, n-ary relations, Representing relations, Equivalence relations, Partial orderings, Hasse diagram, Chains and Antichains, Lattice, Transitive closure and Warshall's algorithm.

Functions: Surjective, Injective and Bijective functions, Identity function, Inverse functions and Compositions of functions.

Module-III	Graph Theory and Applications	9 Hrs.

Graph: Graph Terminology and Special Types of Graphs, Representing Graphs and Graph Isomorphism, Euler and Hamilton Paths, the handshaking lemma, Single source shortest path-Dijkstra's Algorithm, Planar Graphs.

Trees: Trees Introduction, properties of trees, Binary tree, Binary search tree, tree traversal, Expression tree, Huffman coding, Minimum Spanning Tree, Kruskal's and Prim's algorithms.

Module-IV	**Algebraic Structures**	**9 Hrs.**

Groups: Algebraic Systems, Semi Groups, Monoids, Groups, Homomorphism and Normal Subgroups.

Rings: Integral Domains and Fields, Coding Theory.

Text Books:

T1.	C L Liu, D P Mohapatra, *Elements of Discrete Mathematics*, McGraw Hill Education, 2013, 4th Edition, ISBN- 978-1259006395.
T2.	Norman L. Biggs, *Discrete Mathematics*, Oxford University Press, 2nd Edition, 2002, ISBN-978-019850717.

Reference Books:

R1.	Kenneth H. Rosen, *Discrete Mathematics and its Applications*, McGraw Hill Education, 2021, 8th Edition, ISBN-978-9390727353.
R2.	Bernard Kolman, Robert Busby and Sharon C. Ross, *Discrete Mathematical Structures,* Pearson Education India, 2015, 6th Edition, ISBN- 978-9332549593.
R3.	Sriram Pemmaraju, Steven Skiena, *Computational Discrete Mathematics,* Cambridge University Press, 2003, 4th Edition, ISBN- 978-1107262904.

Relevant MOOCs Course (Course name and Weblink)

1. **NPTEL course: Discrete Mathematics,** By Prof. Sudarshan Iyengar, Prof. Anil Shukla, IIT Ropar, **Link:** https://onlinecourses.nptel.ac.in/noc24_cs92/preview
2. **NPTEL course: Discrete Mathematics,** By Minirani, MPSTME, NMIMS Deemed University**, Link:** https://onlinecourses.swayam2.ac.in/cec24_ma18/preview

Relevant Topics for Self-study:

Implementing Set Operations programmatically, Logical equivalence using a truth table, Application of permutation and combination in binomial theorem. Basic principle of pigeonhole, Graph Traversal Algorithm, Travelling Salesperson problem, Abelian group and its examples.

Second Year B. Tech (S. Y B. Tech) AY (2026-27)
Computer Engineering (CE)
[2613PC31]: Data Structures Lab (DSL)

Semester	Credits	Teaching Scheme	Examination Scheme
3	2	P: 4 Hrs./ Week	CIE(TW): 50 Marks ESE(PR): 50 Marks

Prerequisite: Students should have prior knowledge of

- C Programming for Problem Solving (CPPS)
- Object Oriented Programming Using C++ (OOPC)

Course Objectives: The objective of this course is to provide students with

- Understanding the working of Searching & Sorting Algorithms.
- Linked list and types of linked list implementations.
- Use of Hash Table using Hash Functions.
- Applications of Stack and Queue data structures.
- Working of Binary Tree operations.

Course Outcomes: After completing this course, students will be able to

CO1: Select and implement appropriate Searching & Sorting Algorithms for given problem to **achieve** a target Big O time complexity.

CO2: Select appropriate types of linked lists implementations for solving problems to **achieve** a target Big O time complexity.

CO3: Choose and implement suitable Hash Functions for data manipulation in Hash Table to ensure a uniform distribution of keys across the table, maintaining an average search time of O(1).

CO4: Make appropriate use of Stack and Queue data structures to solve arithmetic expression conversion, evaluation & job scheduling problems correctly and efficiently.

CO5: Implement primitive operations of various types of Binary Trees.

List of Assignments

Expt. No.	Problem Statement	CO's
	Preliminary Assignment: Study the key components of the Integrated Development Environment. Practice problems for basic C++ programming	
1.	A customer purchased groceries from a store and wants to verify the prices of specific items in the bill where the items are not in sorted order. Write a C++/Java program to apply any 02 appropriate Searching Algorithms to check the cost of a particular item in the bill. Mention the Time and Space Complexities of Algorithms.	CO1

2.	An E-commerce website has a range of products having Product id, name, manufacturer, price and quality rating out of 5. Write a C++/Java Program to display products, a. In Increasing order of Product id (Use Bubble Sort) b. In Increasing order of Product price (Use Selection Sort) c. In Decreasing order of Product Quality Rating (Use Insertion Sort) OR Write a C++/Java program to display the employee names in the order of their joining year. Analyze and compare the efficiency and suitability of different algorithms in solving the problem. Justify the choice of algorithms.	**CO1**
3.	Consider a telephone book database where the information is stored as the client's name and telephone number. Write a C++/Java program to quickly search the client's telephone number from the database with O(1) complexity. (Hint: Make use of a Hash table and hash function).	**CO3**
4.	Consider a person creating a playlist of media/music files. The person wants to store music_id, title, time_duration for every entry and wants the following functionalities: 1. Able to insert a media file at any position dynamically. 2. Able to delete any entry from the list dynamically. 3. Able to sort on any field (music_id, title,time_duration). 4. Able to play continuous looping fashion. OR There is a student's club named 'Techno-Fun' club where students can get membership on request. Similarly one may cancel the membership of a club. The first node is reserved for the president of the club and the last node is reserved for the secretary of the club. Choose a type of linked list to store student PRN and Name. Write functions to: a) Add members b) show members c) delete members, d) Compute total number of members e) Two linked lists exist for two divisions. Concatenate two lists.	**CO2**
5.	Browser maintains the history which refers to a list of recently visited websites or web pages in a data structure. Write a C++/Java program to store the browser history or navigation information dynamically as page_id and recent time-stamp. Write functions to 1. Visit a specific web page. 2. Navigate forward and backward. 3. Add new web pages. 4. Deleting web pages from existing lists. OR The ticket booking system of the theater has to be implemented using C++/Java program. There are M rows and N seats in each row. A doubly circular linked list has to be maintained to keep track of free seats at rows. Assume some random booking to start with. Use an array to store pointers (Head pointer) to each row. On demand 1. The list of available seats is to be displayed. 2. The seats are to be booked. 3. The booking can be cancelled.	**CO2**

6.	In an application maintaining legal documents which are structured hierarchically with sections, subsections, paragraphs, and sub-paragraphs. Design a program to represent and manage such a document structure. Each node will store either a textual data element (title of section, subsection, paragraph, subparagraph), or a pointer to another linked list (subsections, paragraph, subparagraph)Hint: Use Generalized Linked Lists. A = { S1, S2 {P1, P2, P3 {SP1, SP2}}, S3 {P1.P2}, S4, S5}.	**CO2**
7.	In an expression each opening symbol has a corresponding closing symbol and the pairs of parentheses are properly nested. Write a C++/Java program using Stack to check whether a given expression is well parenthesized as well as the correct arithmetic expression or not. OR The expressions can be represented in infix, prefix and postfix form but the system can evaluate postfix expressions easily. Write C++/Java program for expression conversion as infix to postfix and its evaluation using Stack based on given conditions.	**CO4**
8.	Consider a Pizza parlour where the customers are served from both sides of the waiting line. Choose an appropriate data structure to implement a linear list of elements in which additions and deletions of items may be made at either ends of the list.	**CO4**
9.	**A Mini Project:** (Any one based on Binary Tree, Binary Search Tree and all the data structures learned above) In the memory the files and folders are stored in Hierarchical Manner. Write a C++/Java program to create the hierarchical structure of files and folders using Binary Tree. Construct a Binary Search Tree to perform following operations - i. Insert nodes, ii. Display nodes by Preorder, In-order, Post-order, BFS and DFS. iii. Find number of nodes in longest path from root, iv. Minimum and Maximum data value found in the tree, v. Change a tree so that the roles of the left and right pointers are swapped at every node, vi. Search for a value. Build an Expression Tree using a given infix/postfix expression to store and evaluate a mathematical expression. Building a Huffman tree to assign shorter binary codes (fewer bits) to the most frequently occurring characters and longer codes to the least frequent ones, thereby minimizing the overall size of the data when compressed.	**CO5**

Text Books:

T1. Horowitz and Sahani, *Fundamentals of Data Structures in C++,* University Press, 2007, 2nd Edition, ISBN- 978-07-1678-292-6.

T2. Goodrich, Tamassia, Goldwasser, *Data Structures and Algorithms in Python*, Wiley Publication 2021,1st Edition, ISBN- 978-93-5424-786-6.

T3. Herbert Schildt, *Java: The Complete Reference*, McGraw Hill Education, 2014, 9th Edition, ISBN- 978-0749467241.

Reference Books:

R1. Brassard & Bratley, *Fundamentals of Algorithmic*s, Prentice Hall India/Pearson Education, 1996, 2nd Edition, ISBN- 978-8120311312.

R2. R. Gilberg, B. Forouzan, *Data Structures: A Pseudocode approach with C++*, Cengage Learn, 2005,

2nd Edition, ISBN- 978-8131503140.
R3. M. Weiss, *Data Structures and Algorithm Analysis in C++*, 2nd edition, Pearson Education, 2002, ISBN- 978-0201498400.

Relevant Topics for Self-study: –

Second Year B. Tech (S. Y B. Tech) AY (2026-27)
Computer Engineering (CE)

[2613PC32]: Computer Organization and Architecture Lab (COAL)

Semester	Credit	Teaching Scheme	Examination Scheme
3	1	P: 2 Hrs./ Week	CIE(TW): 25 Marks ESE(OR): 50 Marks

Prerequisite: Students should have prior knowledge of

- Digital Electronics and Logic Design Laboratory

Course Objectives: The objective of this course is to provide students with

- Basic architectural features of processors.
- Instruction set and logic to build assembly language programs.
- Command line argument and stack simulation using tools.
- Multithreading concepts using libraries.

Course Outcomes: After completing this course, students will be able to

CO1: Write and Debug assembly programs using the instruction set to successfully manipulate CPU registers, flags, and memory addresses according to the architectural specifications.

CO2: Examine a function call using a visualization tool to identify how the stack pointer (SP) and base pointer (BP) shift, ensuring the logical order of execution matches the Last-In-First-Out (LIFO) property of th**e** stack.

CO3: Apply multithreading techniques to ensure consistent and thread-safe memory access across all concurrent executions.

List of Assignments

Expt. No.	Problem Statement	CO's
	Preliminary Assignment: Study inline assembly programming in C, data types and memory allocation and deallocation methods.	
1.	Arithmetic operations using inline assembly, a) Write an inline assembly program to add two integer numbers: i. short integers ii. long integers b) Write an inline assembly program to perform signed multiplication and division of two integers.	**CO1**

2.	Write inline assembly code for string and array processing using inline assembly, a) Write an inline assembly program to: i. Accept a string from the user ii. Compute and display the length of the string b) Write an inline assembly program to count and display the number of positive and negative elements in the array	**CO1**
3.	Stack and Control Flow Using Inline Assembly, Simulate stack operations involved in a function call with parameters using an appropriate tool (e.g., debugger / simulator): i. Parameter passing ii. Return address iii. Stack frame creation and deletion	**CO2**
4.	Write an inline assembly program to compute the factorial of a given integer entered through the command line.	**CO2**
5.	A mini project for demonstrating multi-threading using an application with shared memory access (e.g. Arithmetic calculator).	**CO3**

Text Books:

T1. K. R. Irvine, *Assembly Language for x86 Processors*, 8th ed. Boston, MA, USA: Pearson Education, 2019, ISBN-13. 978-0135381694.

T2. J. Cavanagh, *x86 Assembly Language and C Fundamentals*, Boca Raton, FL, USA: CRC Press, 2013, ISBN-13. 978-1466568242

Reference Books:

R1. W. Stallings, *Computer Organization and Architecture: Designing for Performance*, 10th ed. Boston, MA, USA: Pearson, 2016, ISBN-10. 9789332570405.

R2. D. Kusswurm, *Modern x86 Assembly Language Programming: Covers x86-64 SIMD, AVX, and AVX-512*, Berkeley, CA, USA: Apress, 2019, ISBN-13: 978-1484240625.

Relevant Topics for Self-study: -

Second Year B. Tech (S.Y B. Tech) AY (2026-27) Computer Engineering (CE) [2603MD1X]: Multidisciplinary Minor (MDM-1)			
Semester	**Credits**	**Teaching Scheme**	**Examination Scheme**
3	**2**	**L: 2 Hrs./ Week**	**CIE: 20 Marks** **ISE: 20 Marks** **ESE: 60 Marks**
Refer : Multidisciplinary Minor Courses			

Second Year B. Tech (S.Y B. Tech) AY (2026-27) Computer Engineering (CE) [2603MD5X]: Multidisciplinary Minor (MDM-1)			
Semester	**Credits**	**Teaching Scheme**	**Examination Scheme**
3	**1**	**Tut: 1 Hr./ Week**	**CIE (TW): 25 Marks**

Second Year B. Tech (S. Y B. Tech) AY (2026-27)
Computer Engineering (CE)
[2613EE11]: Engineering Economics and Financial Management (EEFM)

Semester	Credits	Teaching Scheme	Examination Scheme
3	1	L: 1 Hr./ Week	CIE(TW): 25 Marks

Prerequisite: Students should have prior knowledge of

- Basic Mathematics, Introductory Engineering Concepts.

Course Objectives: The objective of this course is to provide students with

- Concept of time and value of money.
- Knowledge of financial evaluation techniques.

Course Outcomes: After completing this course, students will be able to:

CO1: Compute the worth of money at various points of time.
CO2: Apply various depreciation methods in determining the value of an asset.
CO3: Evaluate the replacement of an existing asset based on standard replacement analysis techniques.
CO4: Evaluate the best alternative in Engineering Economics problems considering risk and safety.

COURSE CONTENTS

Module-I	Financial & Economic Decision Making Techniques	7 Hrs.

Time value of money; cash flow analysis (single, uniform, gradient); interest rates and factors; present, future, annual, capitalized worth; capital recovery; rate of return; comparison, incremental, and replacement analysis; problems and case studies.

Module-II	Rate of Return, Depreciation & Risk Analysis	6 Hrs.

Break-even analysis; rate-of-return evaluation; depreciation methods (physical/functional, straight line, declining & double declining balance, sum-of-years' digits, sinking fund, service output); financial statements and ratio analysis; project risk, safety, risk–benefit analysis, and risk reduction; case studies

Text Books:

T1.	Chan S. Park, *Contemporary Engineering Economics*, Pearson Prentice Hall, 2007, 4th Edition, ISBN- 978-0131492480.
T2.	Thuesen G. J, *Engineering Economics*, Prentice Hall of India, 2005, 8th Edition, ISBN- 978-0131233232.
T3.	Blank Leland T. and Tarquin Anthony J., *Engineering Economy*, Tata McGraw Hill, 2002, 5th Edition, ISBN- 978-0074620427.

Reference Books:

R1.	Prasanna Chandra, *Fundamentals of Financial Management*, Tata McGraw Hill, 2006, 5th Edition, ISBN- 978-0070609137.
R2.	Mike W. Martin and Roland Schinzinger, *Ethics in Engineering,* Tata McGraw Hill, 2003, 4th Edition, ISBN- 978-0074602751.
R3.	Govindarajan M, Natarajan S, Senthil Kumar V. S, *Engineering Ethics*, Prentice Hall of India,

2004, 1st Edition, ISBN: 978-0131452412.

Relevant MOOCs Course (Course name and Weblink)

- NPTEL course: https://nptel.ac.in/courses/112107209, By Prof. Pradeep K. Jha, IIT Roorkee, Link: https://nptel.ac.in/courses/112107209
- NPTEL course: Financial Management for Managers, By Prof. Anil K. Sharma, IIT Roorkee, Link: https://onlinecourses.nptel.ac.in/noc20_mg31/preview

Relevant Topics for Self-study:

Micro, Small & Medium Enterprises (MSME) and its role in the Indian Economy.

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Second Year B. Tech (S.Y B. Tech) AY (2026-27)

Computer Engineering(CE)

[2603AE31]: Professional Development and Career Readiness (PDCR)

Semester	Credits	Teaching Scheme	Examination Scheme
3	1	P: 2 Hrs./ Week	CIE (TW): 25 Marks

Refer : Common Courses

Second Year B. Tech (S. Y B. Tech) AY (2026-27)

Computer Engineering (CE)

[2603OE51]: OE-I Foreign Language Studies - German (FLSG)

Semester	Credits	Teaching Scheme	Examination Scheme
3	2	L: 2 Hrs./ Week	CIE(TW): 50 Marks

Refer Annexure-II for guidelines

Refer : Common Courses

Second Year B. Tech (S. Y B. Tech) AY (2026-27) Computer Engineering (CE) [2603OE52]: OE-I Foreign Language Studies - Japanese (FLSJ)			
Semester	**Credits**	**Teaching Scheme**	**Examination Scheme**
3	**2**	**L: 2 Hrs./ Week**	**CIE(TW): 50 Marks**
Refer Annexure-II for guidelines			
Refer : Common Courses			

Second Year B. Tech (S. Y B. Tech) AY (2026-27) Computer Engineering (CE) [2603VE11]: Universal Human Values (UHV)			
Semester	**Credits**	**Teaching Scheme**	**Examination Scheme**
3	**2**	**L: 2 Hrs./ Week**	**CIE(TW): 25 Marks**
Refer : Common Courses			

Second Year B. Tech (S.Y B. Tech) AY (2026-27) Computer Engineering(CE) [2603CE31]: Community Engagement Project (CEP)			
Semester	**Credit**	**Teaching Scheme**	**Examination Scheme**
3	**1**	**P: 2 Hrs./ Week**	**CIE (TW): 25 Marks**
Refer : Common Courses			

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Second Year B-Tech
(S. Y. B. Tech)
Semester-4

Second Year B. Tech (S. Y B. Tech) AY (2026-27)
Computer Engineering (CE)

[2614PC11]: Software Engineering (SE)

Semester	Credits	Teaching Scheme	Examination Scheme
4	3	L: 3 Hrs./ Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Fundamentals of Programming Languages

Course Objectives: The objective of this course is to provide students with

- Concepts of software engineering and its diagrams.
- Project planning and estimation.
- Application of software modeling for problem solving.
- Knowledge of risk analysis and software testing techniques.

Course Outcomes: After completing this course, students will be able to:

CO1: Analyse the requirements to model the software requirements through Use Case and Activity diagrams to accurately represent system behaviour and user interactions and produce a Software Requirements Specification (SRS) document that satisfies completeness, consistency, and traceability.

CO2: Calculate project effort and cost using Function Point Analysis or COCOMO II models to estimate the actual resource requirements.

CO3: Design modular and maintainable software systems using appropriate design processes, patterns, and core design concepts by ensuring that the changes in one module have a zero-to-minimal impact on external modules.

CO4: Identify the risk and evaluate the software using software testing by ensuring best pass rate against specified requirements.

COURSE CONTENTS

Module-I	Introduction	12 Hrs.

Introduction to software engineering, The nature of software, Define software, software engineering practice, Software Process: A generic process model, A framework activity, Requirement Vs Analysis Vs Architecture Vs Design Vs Development 4+1 view Architecture, Introduction to UML -Basic building blocks.

Requirement Engineering: Building the requirements model. Agile Model.

Case study: Write Software Requirements Specification (SRS) in IEEE format for given problem statements.

Suggested List of Lab Assignment: Idea Inception for a given problem statement.

Module-II	Planning in Software Engineering	9 Hrs.

Estimation: Project planning process, defining scope, checking feasibility, Resource management, Decomposition techniques, software sizing, LOC based & FP based Estimation. The COCOMO-II Model.

Project Scheduling: Defining the task for project scheduling. **Case Study:** Preparing requirement traceability matrix.		
Module-III	**Modeling in Software Engineering**	**9 Hrs.**
Design: Process, Attributes. Design Patterns: Creational Design Pattern, Structural Design Pattern, Behavioral Design Pattern. Concepts: Abstraction, Architecture, Modularity, information hiding, functional independence, refactoring. **Case study-** Design for a given problem statement.		
Module-IV	**Risk Management and Deployment**	**9 Hrs.**
Risk Management: Software risks, risk identification, risk projection, risk refinement, risk mitigation, monitoring and management (The RMMM plan). Software Configuration Management Testing: verification & validation, software testing strategies. **Case Study:** Selenium testing for given web application.		

Text Books:

T1.	Roger Pressman, *Software Engineering: A Practitioner's Approach*, McGraw Hill, 2009, 7th Edition, ISBN- 007–337597–7.
T2.	Ian Sommerville, *Software Engineering*, Addison and Wesley, 2010, 9th Edition, ISBN-0-13-703515-2.

Reference Books:

R1.	S K Chang, *Handbook of Software Engineering and Knowledge Engineering*, World Scientific, Vol I, II, 2002, 1st Edition, ISBN- 978-981-02-4973-1.
R2.	Pankaj Jalote, *An Integrated Approach to Software Engineering*, Springer, 1997, 2nd Edition, ISBN- 978-8173192715.
R3.	Carlo Ghezzi, *Fundamentals of Software Engineering*, Prentice Hall India, 2002, 2nd Edition, ISBN-10: 0133056996
R4.	Tom Halt, *Handbook of Software Engineering*, Clanrye International, 2015, 1st Edition, ISBN- 10: 1632402939.

Relevant MOOCs Course (Course name and Weblink)

- NPTEL course: Software Engineering By Prof. Rajiv Mall from IIT Kharagpur. Link: https://nptel.ac.in/courses/106105087
- NPTEL course: Software Engineering By Prof. R.Baskaran from IIT Kanpur. Link: https://onlinecourses.swayam2.ac.in/e-learning/preview/cec26_cs06

Relevant Topics for Self-study:

Case Studies for various Applications like: Design a software system for an Automated Teller Machine (ATM) that allows bank customers to perform transactions such as cash withdrawal, balance inquiry, deposit, and PIN change. The system must authenticate users through a PIN, process transactions securely, update account balances in real-time, and provide appropriate messages or error notifications. The ATM should interface with the bank's central system for account verification and transaction processing.

Second Year B. Tech (S. Y B. Tech) AY (2026-27)
Computer Engineering (CE)
[2614PC12]: Database Management System (DMS)

Semester	Credits	Teaching Scheme	Examination Scheme
4	3	L: 3 Hrs./ Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Discrete Mathematics, Data Structures

Course Objectives: The objective of this course is to provide students with

- Fundamental concepts of database management like database design and database languages.
- SQL and NoSQL concepts through database management tools.
- Basics of transaction management and concurrency control.

Course Outcomes: After completing this course, students will be able to:

CO1: Design an Entity-Relationship (E-R) Model and transform it into a relational schema to ensure data integrity and minimize redundancy**.**

CO2: Write SQL queries utilizing joins, subqueries, and aggregate functions to successfully retrieve and manipulate data with complete accuracy in the given result set and PL/SQL blocks including stored procedures, functions, and triggers to automate database tasks while maintaining data integrity.

CO3: Illustrate NoSQL architectural concepts by designing a schema-less document and writing queries for CRUD operations in MongoDB**.**

CO4: Describe the transaction management by explaining how the ACID (Atomicity, Consistency, Isolation, Durability) properties are maintained to ensure data remains valid even in the event of system failures.

COURSE CONTENTS

Module-I	Introduction to Databases and Database Design	12 Hrs.

Introduction: Need for Database Management Systems, Evolution, Database System Concepts,and Architecture.

Data Modeling: Entity Relationship (ER) Model,Relational Model, Extended ER Model, converting ER and EER diagram into tables.

Database Design: Codd's Rules, Need of Normalization, Functional Dependencies, Functional Dependency Closure, Decomposition Properties, Normal Forms: 1NF, 2NF, 3NF, and BCNF.

Module-II	SQL and PLSQL	9 Hrs.

SQL Characteristics and Advantages, Data Types and Literals, DDL, DML, Select Queries and clauses, SQL Operators, Functions, Aggregate Functions, Nested queries, Join Queries,

Views: Creating, Dropping, Updating, Index and Sequence in SQL, DCL, TCL,

PL/SQL: Procedure, Function, Cursors, Trigger.

Module-III	NoSQL Databases	9 Hrs.
Introduction to NoSQL Databases, Types of NoSQL Databases, BASE properties, CAP Theorem, Comparative study of RDBMS and NoSQL, MongoDB (syntax and usage): CRUD Operations, Indexing, Aggregation, MapReduce.		

Module-IV	Database Transactions	9 Hrs.
Basic concepts of a Transaction, ACID Properties, State diagram, Concept of Schedule. **Serializability** – Conflict and View, Concurrency Control Protocols-Lock based and timestamp-based protocols, Recovery techniques.		

Text Books:

T1. Avi Silberschatz, Henry F. Korth, S. Sudarshan, *Database System Concepts*, Tata McGraw-Hill, 2019, 7th Edition, ISBN 9780078022159.

Reference Books:

R1. Shannon Bradshaw, Eoin Brazil, Kristina Chodorow, *MongoDB: The Definitive Guide*, O'Reilly Publications, 2020, 3rd Edition, ISBN- 978-1-449-34468-9.

R2. S. K. Singh, *Database Systems: Concepts, Design and Application*, Pearson Education, 2013, 2nd Edition, ISBN- 978-81-317-6092-5

Relevant MOOCs Course (Course name and Weblink)

- NPTEL course: Fundamentals of Database Systems, By Dr. Arnab Bhattacharya from IIT Kanpur. Link: https://nptel.ac.in/courses/106104135
- NPTEL Course: Database Management System, By Prof. Partha Pratim Das, Prof. Samiran Chattopadhyay from IIT Kharagpur. Link: https://onlinecourses.nptel.ac.in/noc26_cs39/preview

Other Resources/Links:

- https://www.opensourceforu.com/

Relevant Topics for Self-study:

Inference Rules, Advanced nested queries in SQL.

Second Year B. Tech (S. Y B. Tech) AY (2026-27)
Computer Engineering (CE)
[2614PC13]: Operating Systems (OS)

Semester	Credits	Teaching Scheme	Examination Scheme
4	**3**	**L: 3 Hrs./ Week**	**CIE: 20 Marks** **ISE: 20 Marks** **ESE: 60 Marks**

Prerequisite: Students should have prior knowledge of
- Computer Fundamentals

Course Objectives: The objective of this course is to provide students with
- OS internals and functioning.
- Organization of data and I/O devices in OS.
- Mechanism of interacting user space applications with kernel space, Linux system calls.
- Shell scripting constructs.

Course Outcomes: After completing this course, students will be able to:

CO1: Differentiate various process scheduling and memory allocation algorithms under varying scenarios based on predefined performance metrics (waiting time, turnaround time, throughput and response time).

CO2: Compare disk scheduling algorithms and file allocation methods, justifying the selection of a specific method based on its ability to minimize seek time and fragmentation.

C**O3: Design** applications using appropriate system calls, such as process and file, to solve real-world scenarios, and evaluate their correctness, efficiency, and resource utilisation.

CO4: Write shell scripts to automate user account management and resource monitoring, ensuring scripts execute without syntax errors and adhere to access policies**.**

COURSE CONTENTS

Module-I	Introduction, Process and Memory Management	12 Hrs.

Introduction: Introduction to Operating Systems, definition,OS Services, and types of operating systems, OS Design: Monolithic, Microkernel, and Layered systems.

Process Management: Process Concept: Process states and Process Control Block (PCB), CPU **Scheduling:** Scheduling criteria, types of schedulers, Scheduling algorithms: FCFS, SJF, Priority, and Round-Robin, Process Synchronization: Critical Section Problem, Deadlock: Conditions, Prevention, and Detection. Recovery. Inter-process Communication (IPC) : message passing.

Memory Management: Memory Management requirements (relocation and protection), Memory partitioning, Placement algorithms: First Fit, Best Fit, and Worst Fit, Virtual Memory and Demand Paging, Paging and page tables, Page replacement algorithms: FIFO, LRU, and Optimal, Segmentation and comparison with paging.

Module-II	File Systems and Device Management	9 Hrs.

File Systems: Overview of File Systems, File organization and access methods, File directories, File sharing, Secondary storage management, File system security, UNIX File System, Linux Virtual File System, and

Device Management: I/O devices, Organization of the I/O function, Operating system design issues for I/O, I/O buffering, Disk scheduling policies: FIFO, SSTF, SCAN, and C-SCAN, Linux I/O.
Case Study : Windows File System and I/O.

Module-III	Systems Calls Design and Implementation	9 Hrs.

Introduction to System Calls: Definition of system calls and their role in the OS, Differences between user mode and kernel mode, Overview of the system call interface.
Mechanisms of System Calls: Working of a system call (traps, interrupts),Transition from user mode to kernel mode, Context switching and its overhead.
Types of System Calls: Process Control: Creating and terminating processes (fork, exec, exit), Process synchronization (wait, signal), Process scheduling and priority management.
File: File creation, deletion, and manipulation (open, read, write, close), File attributes and permissions, Directory operations (mkdir, rmdir,chdir).
Error Handling: Common errors encountered with system calls, Error codes and how to interpret them, Best practices for robust error handling.

Module-IV	Shell Scripting and Modern OS	9 Hrs.

Introduction to Shell Scripting: Overview of shell scripting and its purpose,Types of shells (e.g., Bash, Zsh, Ksh, Csh), Differences between command line and shell scripts.Basic Shell Commands.
Writing Shell Script: Script structure and syntax,Creating and executing a simple shell script, Using the shebang (#/!) for specifying the interpreter. User management.Variables and Data Types, Control Structures, Functions, Input and Output
Modern Operating Systems and concepts, virtualization, containerization, and Docker.

Text Books:

T1.	Abraham Silberschatz, Peter Baer Galvin, Greg Gagne, *Operating System Principles*, 2006, 8th Edition, Wiley India Private Limited, ISBN: 9788126509621.
T2.	William Stallings, *Operating Systems: Internals and Design Principles,* Pearson Education, 2006, 9th Edition, ISBN: 978-9352866717.
T3.	Maurice J. Bach, "The Design of the UNIX Operating System", Prentice-Hall, 1986, 8th Edition, ISSN: 978-0132017991.

Reference Books:

R1.	Andrew S. Tanenbaum , *Modern Operating Systems*, Prentice Hall of India, 2009, 3rd Edition, ISBN- 978-8120339040.
R2.	Deitel & Deitel , *Operating systems,* Pearson Education, 2003, 3rd Edition, ISBN-978-0131828278.

Relevant MOOCs Course (Course name and Weblink)

1. NPTEL course: [Operating Systems Fundamentals](https://onlinecourses.nptel.ac.in/noc24_cs108/preview), By Prof. Santanu Chattopadhyay, IIT Kharagpur. Link : https://onlinecourses.nptel.ac.in/noc24_cs108/preview

Other Resources/Links: https://www.opensourceforu.com/

Relevant Topics for Self-study:

I/O management, Interrupt handling, Device drivers, Daemons, Dynamic Link Libraries concepts in Windows/Ubuntu/ Android, etc.

Second Year B. Tech (S. Y B. Tech) AY (2026-27)
Computer Engineering (CE)

[2614PC31]: Operating Systems Lab (OSL)

Semester	Credits	Teaching Scheme		Examination Scheme
4	1	P: 2 Hrs./ Week		CIE(TW): 50 Marks ESE (PR): 25 Marks

Prerequisite: Students should have prior knowledge of

- Fundamental Programming.

Course Objectives: The objective of this course is to provide students with

- Process and memory management techniques.
- Disk scheduling and internals of storage management.
- Usage of various system calls and shell scripting.

Course Outcomes: After completing this course, students will be able to

CO1: Execute CPU scheduling algorithms (FCFS, SJF, RR, Priority) in simulated scenarios, with assessment based on accurate derivation and comparison of performance metrics such as waiting time, turnaround time, response time, and CPU utilisation**.**

CO2: Implement different page replacement strategies (FIFO,LRU,Optimal) in a simulated OS environment with varying process sizes, and evaluate efficiency using performance metrics such as page faults.

CO3: Demonstrate Linux file system internals in a practical environment, and evaluate functionality using metrics such as access time and storage efficiency**.**

CO4: Write shell scripts to manage users and access rights in a Linux environment, and assess effectiveness through correctness of access control and execution efficiency.

List of Assignments

Expt. No.	Problem Statement	CO's
	Preliminary Assignment: Study the Linux OS and Shell (as command line interface.) What is kernel? How to find kernel version. **Basic Shell Commands**: Common Unix/Linux commands (ls, cp, mv, rm, cat,echo),	
1.	Given real-world scenarios such as **Banking System, Batch Processing System, Online Ticket Reservation System, and Emergency Services**, identify a suitable CPU scheduling algorithm-FCFS, Shortest Job First(SJF), Priority, Round Robin for each case and implement the chosen algorithm(s) using **C / C++ / Java / Python**. The program shall accept the number **of processes, arrival time, burst time, priority, and time quantum (as applicable)** as input, and compute the **average waiting time** and **average turnaround time**.	**CO1**
2.	Write a program to simulate Page replacement strategies of virtual memory management : First in Firstout, Least recently used and Optimal page replacement.	**CO2**

3.	Write a program demonstrating Linux file system internals by using file and directory related system call to create, access, modify, and manage files and directories along with appropriate error handling	**CO3**
4.	Implement shell-based automation to manage users, groups, and file access rights in a Linux operating system using shell scripting constructs and commands.	**CO4**
5.	**Case study**: (Any One) 1)Developing an application and containerization of Application. 2) Linux Boot Loaders.	**CO1-4**

Text Books:

T1.	Abraham Silberschatz, Peter Baer Galvin, Greg Gagne, *Operating System Principles*, 2006, 8th Edition, Wiley India Private Limited, ISBN: 9788126509621.
T2.	William Stallings, *Operating Systems: Internals and Design Principles*, Pearson Education, 2006, 9th Edition, ISBN: 978-9352866717.
T3.	Maurice J. Bach, *The Design of the UNIX Operating System*, Prentice-Hall, 1986, 8th Edition, ISSN: 978-0132017991.

Reference Books:

R1.	Andrew S. Tanenbaum , *Modern Operating Systems*, Prentice Hall of India, 2009, 3rd Edition, ISBN- 978-8120339040.
R2.	Deitel & Deitel , *Operating systems*, Pearson Education, 2003, 3rd Edition, ISBN-978-0131828278.

Relevant MOOCs Course (Course name and Weblink)

1. NPTEL course: Operating Systems Fundamentals, By Prof. Santanu Chattopadhyay, IIT Kharagpur. Link : https://onlinecourses.nptel.ac.in/noc24_cs108/preview

Other Resources/Links:

https://www.opensourceforu.com/

Relevant Topics for Self-study:

I/O management, Interrupt handling, Device drivers, Daemons, Dynamic Link Libraries concepts in Windows/Ubuntu/ Android, etc.

Second Year B. Tech (S. Y B. Tech) AY (2026-27)
Computer Engineering (CE)
[2614PC32]: Database Management System Lab (DMSL)

Semester	Credits	Teaching Scheme	Examination Scheme
4	2	P: 4 Hrs./ Week	CIE(TW): 50 Marks ESE (PR): 25 Marks

Prerequisite: Students should have prior knowledge of

- Basic Programming

Course Objectives: The objective of this course is to provide students with

- Knowledge to design ER model, handle SQL statements and PL/SQL code in databases.
- Skills to handle NoSQL databases.
- Application development with database connectivity.

Course Outcomes: After completing this course, students will be able to

CO1: Design an E-R model for given requirements using ERD Plus/ERWin tool and convert the same into database tables and apply normalization.

CO2: Implement the given relational schema, database design using ER Model, queries, and PL/SQL programs for 2-tier architecture using MySQL

CO3: Implement NoSQL queries and aggregate functions for given requirements using MongoDB.

CO4: Develop database applications using database connectivity.

List of Assignments

Expt. No.	Problem Statement	CO's
	Preliminary Assignment: Study of database design along with identification of suitable attributes, attribute types, and constraints.	
1.	Develop an Entity Relationship Model using open source tools for a real world problem statement.Convert the ER diagram into relational tables and apply the normalization for the same.Use the same problem statement for the minproject implementation,	**CO1**
2.	Use an open source database tool to implement a structured database system for a Retail Store to manage products, customers and orders. Write SQL queries to perform DDL operations for table creation in the Retail Store database with constraints (Not Null, Primary, Foreign etc) and Modification using Alter command variations, and also perform basic DML operations (Insert, Select, Update, Delete), and executing various Select statement queries with different clauses for given schemas of Retail Store database.	**CO2**
3.	An University maintains a structured database to manage student enrollments, courses, instructors and departments. Write SQL queries using various JOIN types (Natural, Inner, Equi, Non-Equi, Outer, Left Outer, Right Outer), as well as Subqueries and Views to extract insights from the university database system.	**CO2**

4.	A Bank wants to automate its customer credit scoring system based on their financial standing. Let the credit score be computed on a scale of 0 to 5 by a PLSQL function based on the account balance and loan amount of any customer. (hint: if loan_amount is 50% of account balance then, credit_score: 3). Write a stored procedure that uses a cursor and calls the function with each customer name for computing credit scores of all bank customers. i. Customer(Cust_name, AccNo, Balance, city) ii. Loan(Loan_no, branch_name, Amount) iii. Borrower(Cust_name, Loan_no, CreditScore)	**CO2**
5.	A Company wants to track employee salary changes, maintain company-wide statistics, and log employment history efficiently using row-level triggers. Implement BEFORE and AFTER triggers on EMPLOYEE, COMPANY_INFO, and EMP_LOG tables using INSERT, UPDATE and DELETE operations. i. EMPLOYEE(Emp_Id, First_Name, Last_Name, Email, Phone_No, Hire_Date, Job_Profile, Salary, HRA) ii. COMPANY_INFO(Emp_Count, Total_Salary_Expenses) iii. EMP_LOG(Emp_Id, Old_Salary, New_Salary, Edit_Time, Job_Status)	**CO2**
6.	An institute maintains details of all teachers, including name, qualifications, department details, experience, salary structure, date of joining, appointement_nature and area of expertise. Design and implement MongoDB queries to perform CRUD operations on the teachers collection for various administrative tasks. Create the above collection, insert suitable documents and design updation and retrieval queries requiring comparison and logical operators, save() method, etc.	**CO3**
7.	Customers of an online marketplace frequently search for products and their orders, and the search becomes slow as the products and orders grow. Implement all types of indexes on the products and order MongoDB collections.	**CO3**
8.	The University wants to analyze course enrollments, faculty performance and student achievements using MongoDB aggregation pipelines. The university maintains a “courses” collection, which includes: i. Course details (title, department, credit hours, instructor) ii. Student enrollments (student names, scores, pass/fail status) iii. Faculty details (experience, designation, department) Design and implement aggregation queries to generate various reports using multiple aggregation stages with the given collection in MongoDB.	**CO3**
9.	MiniProject- Use the problem statement selected in assignment-1one and design a database application using database connectivity. Students may select the following requirements. Front End: Java/P/PHP/Python/Ruby/.net/any other languageBackend: MongoDB/ MySQL.	**CO3**
Text Books:		

T1.	Avi Silberschatz, Henry F. Korth, S. Sudarshan, *Database System Concepts*, Tata McGraw-Hill, 2019, 7th Edition, ISBN 9780078022159.	
Reference Books:		
R1.	Shannon Bradshaw, Eoin Brazil, Kristina Chodorow, *MongoDB: The Definitive Guide*, O'Reilly Publications, 2020, 3rd Edition, ISBN- 978-1-449-34468-9.	
R2.	S. K. Singh, *Database Systems: Concepts, Design and Application*, Pearson Education, 2013, 2nd Edition, ISBN- 978-81-317-6092-5	
Other References		
1.	https://dev.mysql.com/doc/refman/8.4/en/tutorial.html	
Relevant Topics for Self-study:		
Advanced nested queries in SQL		

Second Year B. Tech (S. Y B. Tech) AY (2026-27) Computer Engineering (CE) [2604MD1X]: Multidisciplinary Minor (MDM-2)			
Semester	**Credits**	**Teaching Scheme**	**Examination Scheme**
4	2	P: 2 Hrs./ Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks
Refer : Multidisciplinary Minor Course			

Second Year B. Tech (S. Y B. Tech) AY (2026-27) Computer Engineering (CE) [2604MD3X]: Multidisciplinary Minor (MDM-2)			
Semester	**Credits**	**Teaching Scheme**	**Examination Scheme**
4	1	P: 2 Hrs./ Week	CIE(TW): 25 Marks
Refer : Multidisciplinary Minor Course			

Second Year B. Tech (S. Y B. Tech) AY (2026-27) Computer Engineering (CE) [2604OE51]: OPEN ELECTIVE -II [NPTEL / SWAYAM / MOOCs] (OE-II)			
Semester	**Credits**	**Teaching Scheme**	**Examination Scheme**
4	2	P: 2 Hrs./ Week	ESE: 50 Marks
Refer : Annexure-II for guidelines			

Second Year B. Tech (S. Y B. Tech) AY (2026-27)
Computer Engineering (CE)
[2614VS31]: Project-Based Learning (PBL)

Semester	Credits	Teaching Scheme	Examination Scheme
4	2	P: 4 Hrs./ Week	CIE(TW): 50 Marks ESE (OR): 25 Marks

Prerequisite: Students should have prior knowledge of

- Basic Subject Knowledge: Foundational understanding of the technology stack.
- Design thinking lab

Course Objectives: The objective of this course is to provide students with

- Independent learning by multi-disciplinary problem-solving with social context.
- Opportunity to get involved either individually or as a group so as to develop team skills and learn professionalism.

Course Outcomes: After completing this course, students will be able to

CO1: Explain the project lifecycle for a real-world or societal problem with stakeholder alignment**.**

CO2: Apply design and development skills to implement a functional project solution with successful integrated testing**.**

CO3: Analyse and decompose complex problems into modular components for efficient development and maintenance**.**

CO4: Evaluate project methods, tools, and processes for effectiveness and quality**.**

CO5: Design, develop, and test a comprehensive project solution addressing a societal need with stakeholder validation**.**

List of Assignments

Expt. No.	Problem Statement	CO's
1.	Group formation and project Ideation domain wise (please refer the attached list of sample project ideas but not limited to the list)	**CO1, CO3**
2.	**Project Initialization & Requirements Gathering** 1. Define the problem statement and project goals. 2. Collect functional and non-functional requirements from stakeholders. Conduct a feasibility study to ensure the project is realistic in terms of resources, cost, time, and skills.	**CO1, CO3**
3.	**Technology Stack Selection** 1. Choose the programming languages, frameworks, tools, and technologies required for the project. Set up version control (Git), development environment (IDEs), and testing frameworks.	**CO2, CO3**
4.	**System Design & Architecture** 2. Design the system architecture (e.g., client-server, database	

	schema, components etc). 3. Create wireframes or UI/UX prototypes. 4. Review design with stakeholders or faculty to refine and ensure alignment with project goals. Algorithm/Flow Chart, ER Diagram and Use Case Diagrams	**CO1, CO2**
5.	**Finalize Design & Documentation (SRS)** 1. Finalize design and document architecture, components, and UI flow. Plan a detailed project timeline and assign roles.	**CO1, CO3**
6.	**Frontend and Backend Development** 1. Designing and implementing user interfaces (UI). 2. Handling user interactions and dynamically updating the UI with data fetched from the backend using APIs. 3. Creating and managing databases (using SQL or NoSQL databases). 4. Developing APIs to handle requests from the frontend. 5. Implementing business logic, authentication, and authorization. Ensuring application security, performance, and scalability.	**CO2**
7.	**Environment Setup & Testing with Feature Development** 1. Set up version control (Git), IDEs, and testing tools. 2. Start coding core features (authentication, main functionality). 3. Implement basic unit/ white box/black box tests for components. Conduct peer code reviews and fix issues.	**CO4, CO5**
8.	**Documentation & Final Refinements** 4. Finalize project documentation (user manuals, technical specs). 5. Address final bugs and refine the codebase. Prepare a video clip of the final running project.	**CO4, CO5**
9.	**Demonstration and presentation of projects**	**CO1, CO2,**
10.	**Research paper writing(if interested)**	**CO3, CO4**

Indicative/ Sample List of Tools (Students are allowed to use any latest tools / Not Limited to this):

1. Git/GitHub - Version control and collaboration.
2. Visual Studio Code - Integrated development environment (IDE).
3. AWS/Google Cloud/Azure - Cloud computing platforms.
4. TensorFlow/PyTorch/Scikit-learn - Machine learning libraries.
5. React/Angular/Django - Web development frameworks.

Text Books:

T1.	Michael E. Auer, *Project-Based Learning in the Computer Science Classroom*, 2019, 1st Edition, ISBN- 978-3-319-70663-0.
T2.	Richard DuFour, Rebecca DuFour, and Robert Eaker, *Learning by Doing: A Handbook for Professional Learning Communities at Work*, Solution Tree, 2006, 1st Edition, ISBN-978-1-4166-0732-8
T3.	Gary R. Morrison, Steven M. Ross, and Jerrold E. Kemp, *Designing Effective Instruction*, John Wiley & Sons, 2010, 6th Edition, ISBN- 978-1-118-20763-4.

T4.	Nell Dale and John Lewis, *Computer Science Illuminated*, Jones & Bartlett Learning, 2019, 7th Edition ISBN- 978-1-118-45306-7.
Reference Books:	
R1.	A. K. Singh, *Microcontrollers: Principles and Applications*. New Delhi, India: Prentice Hall, 2012.
R2.	J. W. Valvano, *Embedded Systems: Introduction to ARM Cortex-M Microcontrollers*, 5th ed. CreateSpace Independent Publishing Platform, 2012.
Other References:	
1.	Junior, E., et al. *Systematic literature review of Gamification and Game-based Learning in the context of Problem and Project Based Learning approaches*. International Symposium on Project Approaches in Engineering Education. Vol. 9. 2019
2.	Malik, Khalid Mahmood, and Meina Zhu. *Do project-based learning, hands-on activities, and flipped teaching enhance student's learning of introductory theoretical computing classes.* Education and information technologies 28.3 (2023): 3581-3604.
3.	Faizi, Jamilurahman, and Mohammad Sarosh Umar. *A conceptual framework for software engineering education: project based learning approach integrated with industrial collaboration.* International Journal of Education and Management Engineering 11.5 (2021): 46.
Relevant Topics for Self-study:	
-	

Suggested Problem Statements (Not limited to this choose any)		
Sr No	Domain	Sample Problem Statement
1	Waste Management Optimization	Inefficient waste collection leads to overflowing bins and increased pollution.
2	Renewable Energy Monitoring	Limited visibility on renewable energy production affects efficiency.
3	Urban Air Quality Monitoring	Poor air quality in urban areas poses health risks.
4	Water Quality Testing	Contaminated water sources threaten public health.
5	Smart Traffic Lights	Traffic congestion causes delays and pollution.
6	Elderly Care Monitoring System	Elderly individuals living alone face health risks and emergencies.
7	Food Safety Tracking	Problem: Foodborne illnesses due to unsafe food handling practices.
8	Energy Consumption Management	Problem: High energy consumption leads to increased costs and carbon footprint.
9	Disaster Response App	Problem: Ineffective communication during natural disasters.
10	Personalized Learning Tools	Problem: One-size-fits-all education fails to meet diverse student needs.

Second Year B. Tech (S. Y B. Tech) AY (2026-27)
Computer Engineering (CE)
[2614EE11]: Entrepreneurship (EP)

Semester	Credit	Teaching Scheme	Examination Scheme
4	1	L: 1 Hr./ Week	CIE(TW): 25 Marks

Prerequisite: Students should have prior knowledge of

- Skills to identify target customers, segment markets.
- Utilization of tools for designing and developing effective business models.
- Understanding of team dynamics and core business principles.

Course Objectives: The objective of this course is to provide students with

- Basic techniques of algorithm performance analysis.
- Knowledge of various data searching and sorting methods with pros and cons.
- Structural constraints and advantages in usage of the data using Hashing.
- Data manipulation using various data structures like Linked List, Stack, Queues.
- Data handling with tree data Structures.

Course Outcomes:
At the end of the course, students will be able to

CO1. Design innovative and sustainable business models.
CO2. Develop business strategies to foster business development.

COURSE CONTENTS

Module-I	Entrepreneurship in Practice	8 Hrs.

Team building, Shared leadership, role of good team, Collaboration tools and techniques, Marketing and sales, Positioning, Channels and strategies, Sales planning, Support, Project management, Planning and tracking, Business Regulation. Customer Segmentation.

Module-II	Economics and Management for Entrepreneurs	7 Hrs.

Customer identification, Market, Creative solution, Unique Value proposition Business Model Canvas, Types of business models, Risk identification, Business Plan documentation, Business Establishment Requirements.

Text Books:

T1.	Robert D. Hisrich, *Advanced Introduction to Entrepreneurship*, Elgar Advanced Introductions series, Extent: 296 pp, 2014, 1st Edition, ISBN- 978- 1 78254 6153.
T2.	Henry Rwigema and Robert Venter, *Advanced Entrepreneurship*, Oxford University Press, 2004, 1st Edition, ISBN- 978- 0195780574.

Reference Books:

R1.	Robert D. Hisrich, Michael P. Peters, Dean A. Shepherd, Sabyasachi Sinha, *Entrepreneurship*, McGraw Hill Education 2020, 11th Edition, ISSN- 978-9390113309.
R2.	Barbara J. Orser and Catherine J. Elliott, *Feminine Capital: Unlocking the Power of Women Entrepreneurs,* Stanford University Press, 2015, 1st Edition, ISBN- 978-0804794312.

Relevant Topics for Self-study:

Case studies on Leadership and Market strategies.

Second Year B. Tech (S.Y B. Tech) AY (2026-27) Computer Engineering(CE) [2604AE31]: Collaborative Skills, Digital Ethics, and Cyber Security (CDC)			
Semester	**Credit**	**Teaching Scheme**	**Examination Scheme**
4	**1**	**P: 2 Hrs./ Week**	**CIE (TW): 25 Marks**
Refer : Common Courses			

Second Year B. Tech (S. Y B. Tech) AY (2026-27) Computer Engineering (CE) [2604VE11]: Indian Constitution and Social Responsibility (ICSR)			
Semester	**Credit**	**Teaching Scheme**	**Examination Scheme**
4	**1**	**L: 1 Hr./ Week**	**CIE(TW): 25 Marks**
Refer : Common Courses			

Annexure-I

Structure of Multi-Disciplinary Minor Courses

The structure for the multidisciplinary Minor courses is as follows.

			Teaching Scheme (Hours/Week)				Credits				Examination Scheme and Marks						
											Theory			Practical			
Sem	Course code	Name of the Course (Short forms)	L	P	T	Total	L	P	T	Total	CIE [20]	ISE [20]	ESE [60]	CIE TW	ESE (PR)	ESE (OR)	Total
3	2603051X1	MDM-1	2	-	-	2	2	-	-	2	20	20	60	-	-	-	100
3	2603052X1	MDM-1 Tut	-	-	1	1	-	-	1	1	-	-	-	25	-	-	25
4	2604051X2	MDM-2	2	-	-	2	2	-	-	2	20	20	60	-	-	-	100
4	2604052X2	MDM-2 Lab	-	2	-	2	-	1	-	1	-	-	-	25	-	-	25
5	2605051X3	MDM-3	2	-	-	2	2	-	-	2	20	20	60	-	-	-	100
5	2605052X3	MDM-3 Lab	-	2	-	2	-	1	-	1	-	-	-	25	-	-	25
6	2606051X4	MDM-4	2	-	-	2	2	-	-	2	20	20	60	-	-	-	100
6	2606052X4	MDM-4 lab	-	2	-	2	-	1	-	1	-	-	-	25	-	-	25
8	2608053X5	MDM-5	-	-	2	2	-	-	2	2	-	-	-	50	-	-	50
		Total	8	6	3	17	8	3	3	14	80	80	240	150	0	0	550

Note: In course code X is basket number. #: is laboratory or tutorial as per course requirements.

1. Students are instructed to select one basket labeled as MD-1 to MD-9 based on applicable criteria in the last column of the table below.
2. Follow the registration process initiated by the institute/ department.
3. Once the course is registered, change of the basket/course in any of the further semesters is not allowed.
4. The total credits will be the same for all the courses except MDM-5.
5. Departments may conduct MDM-5 in virtual mode for the smooth conduction of internship for the full duration.

List of Multi-Disciplinary Minor Domains

Sr. No.	Multi-Disciplinary Minor Domains	SY		TY		B-Tech	Offered to students of BTech Program
		MDM-1	MDM-2	MDM-3	MDM-4	MDM-5	
		Sem-III	Sem-IV	Sem-V	Sem-VI	Sem-VII/VIII	
MD 1	**Smart and Sustainable Systems (SSS)**	Fundamentals of Smart and Sustainable Systems (FS&SS) Tut.	IoT for Smart and Sustainable Systems (ISSS) & Lab	Data Analytics for Smart and Sustainable Systems (DASSS) & Lab	Security for Smart and Sustainable Systems (SSS) & Smart and Sustainable Systems Development (SSD) Lab	Smart and Sustainable System (SSD) (MOOC)	**ALL**
MD 2	**Financial Technology and Management (FTM)**	Finance and Management (FM)	Fundamentals of Financial Engineering (FFE) & Tut	Banking, Financial Services and Insurance (BFSI) & Tut	Fundamentals of Stock Market (FSM) & Tut	Fintech: Foundations & Applications (FFA) & Tut	**ALL**
MD 3	**3D- Printing (3DP)**	3D- Printing (3DP)	3D modeling and Design (3MD) & Lab	Fundamentals of Additive Manufacturing (FAM)& Lab	3D Printing Materials and Processes (3DPMP)	Industry 4.0 and Digital Manufacturing (IDM)	**ALL**
MD 4	**Electric Vehicles (EV)**	Electric Vehicles (EV)	EV foundation – Principles and Concepts (EVPC) & Lab	Advanced Motor Technologies and Power Electronics for EV(AMT) & Lab	EV Powertrain in Dynamics and Control System(PDC) Tut/Lab	Intelligent EV Systems: AI IoT and Automation(IEV)	**ALL**
MD 5	**Applied Mathematics for Engineering (AME)**	Applied Mathematics for Engineering (AME)	Linear Algebra with Python & Lab	Statistical Techniques and Numerical Methods with R & Lab	Fuzzy Logic and Graph Theory with Matlab/Python & Lab	Optimization Techniques & Lab	**ALL**

MD 6	**Software Development (SD)**	Software Development (SD)	Data Structures and Algorithms (DSA) & Lab	Object Oriented Programming (OOP) &Lab	Database and Management Systems (DBMS) & Lab	Web Development (WD) & Lab	**E&TCE**
MD 7	**Autonomous and Intelligent Systems (AIS)**	Autonomous and Intelligent Systems (AIS)	Digital Systems and Organization (DSO) & Lab	Smart System Engineering (SSE) & Lab	Embedded IoT Systems (EIS) & Lab	Autonomous Systems (AS) & Lab	**All except E&TCE**
MD 8	**Embedded Systems- (ES)**	Embedded Systems (ES)	Fundamental of Microcontroller (FM) & Lab	Embedded Processors –I (EP -I) & Lab	Microcontrollers and IoT (MI) & Lab	Embedded Systems and RTOS (ES-RTOS) & Lab	**All Except E&TCE**
MD 9	**AI & Machine Learning (AI-ML)**	AI & Machine Learning (AI-ML)	Statistical Data Analysis & Lab	Machine Learning (ML) & Lab	Natural Language Processing (NLP) & Lab	Artificial Intelligence (AI) & Lab	**E&CE**

Link: [Detailed Syllabus]()

Annexure -2

Guidelines for Open elective Courses

- Open Elective – I will be offered in the third semester as foreign language as prescribed in the structure.
- Open Electives – II, III, IV will be offered through SWAYAM/NPTEL MOOCs of Equivalent Credits.
- Departments will prepare the baskets/list of courses to be offered at the start of every semester, Students may choose any one course from the basket with adhering to any one stream.
- The department shall appoint a faculty mentor to ensure monitoring and course registration smoothly. The load for faculty should be reflected in the time table.
- Equivalent Teaching Scheme, credits and examination schemes are detailed in table below.
- End semester Evaluation: No separate examination is conducted by the institute.

			Teaching Scheme (Hours/Week)				Credits				Examination Scheme and Marks						
Sem	Course Code	Name of the Course	L	P	T	Total	L	P	T	Total	Theory			Practical			Total
											CIE	SE	ESE	CIE	ESE		
											20	20	60	TW	PR	OR	
3	**OE-I**	Foreign Language Studies (FLS)	-	-	2	2	-	-	2	2	-	-	-	50	-	-	50
4	**OE-II**	MOOCs	-	-	2	2	-	-	2	2	-	-	50	-	-	-	50
5	**OE-III**	MOOCs	-	-	2	2	-	-	2	2	-	-	50	-	-	-	50
6	**OE-IV**	MOOCs	-	-	2	2	-	-	2	2	-	-	50	-	-	-	50

Guidelines for MOOCs

- The department shall release a list of approved SWAYAM-NPTEL courses before the commencement of every semester.
- Students shall register for the approved Courses as per the schedule announced by SWAYAM-NPTEL.
- A student shall undergo the courses only from the list notified by the department through SWAYAM/NPTEL platform and complete all the assignments and examination requirements as specified by SWAYAM/NPTEL.
- SWAYAM-NPTEL Courses are considered for transfer of credits only if the student concerned has successfully completed and obtained the SWAYAM-NPTEL Certificate.
- The credit equivalence for SWAYAM-NPTEL Courses: 12 weeks – 3 credits; 8 weeks – 2 credits; 4 weeks – 1 credit.
- Equivalent marks will be considered for awarding the grades as specified in examination rules and regulations. The weightage for assignments is 40%, while the weightage for the proctored examination will be 60% for award calculating SGPA/CGPA. Students must score a minimum of 40% of the total marks by combining both assignments and proctored examinations
- A student must submit the original SWAYAM-NPTEL Course Certificates to the Head of the Department concerned, with a written request for the transfer of the equivalent credits. On verification of the SWAYAM-NPTEL Course Certificates and approval by the head of the department, credits will be awarded.
- The Institute shall not reimburse any fees/expenses a student may incur for the SWAYAM-NPTEL Courses.
- If the SWAYAM/NPTEL course calendar does not align with the institute's calendar, the department shall facilitate and conduct examination of the relevant course of equivalent credits in physical/virtual mode and award the credits accordingly.

Guidelines for Evaluation

Continuous Internal Evaluation: (Weightage for Attendance: 5, Activity Based Learning Evaluation: 15)

The department shall declare the set of all applicable activities such as Problem Based Learning, Quizzes, Small Project, field work, group discussion, but not limited to etc. The course coordinator, in consultation with course teachers, shall select any two activities suitable for the course from the list declared by the department and get the selected activities approved from HoD. The Course teacher shall get the activities carried out by students, evaluate the student performance based on the prescribed rubrics. The department shall prepare the rubrics for all the activities and display the same before the commencement of academics.

In-Semester Examination: Written examination shall be conducted for one hour duration on First Module for 20 marks. End-Semester Examination: Written examination shall be conducted for two- and half-hour duration on Modules II, III, IV for 60 marks.

Annexure -3

Relevant topics for self study:

These topics are optional and suggested for students for self study to get additional exposure. However, these topics will not be considered for any evaluation and credits.

Annexure -4

Contents

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Second Year B-Tech (S. Y B. Tech)

Semester-3

[2603OE5X] Foreign Language Studies (FLS)

Choose any one course from the following courses and report that to department

Second Year B. Tech (S. Y B. Tech) AY (2026-27)
Common to all Programs

[2603OE51]: Foreign Language Studies - German (FLSG)

Semester	Credits	Teaching Scheme	Examination Scheme
3	2	L: 2 Hrs./ Week	CIE(TW): 50 Marks

Prerequisite: Nil

Course Objectives: The objective of this course is to provide students with
- Communicate about everyday topics in German.
- Learn basic German grammar rules.
- Build a practical German vocabulary.
- Gain awareness of German culture.

Course Outcomes: After completing this course, students will be able to

CO1: Introduce themselves and others in German.
CO2: Describe daily life and their surroundings
CO3: Discuss time, jobs, and health in German.
CO4: Plan leisure activities and travel in German

COURSE CONTENTS

Module-I	Introduction, Personal Information, and Basic Grammar	7 Hrs.

Themes:
1. Introducing oneself and others
2. Hobbies
3. Days of the week, months, seasons

Grammar:
1. W questions
2. Personal pronouns
3. Simple sentences
4. Verb conjugation
5. Articles (definite and indefinite)
6. Plurals
7. Verbs "to have" and "to be"

Module Content:
1. Introduction to German greetings and how to introduce oneself.
2. Practicing conjugation of common verbs.
3. Learning W-questions and using personal pronouns in conversation.
4. Discussing hobbies and daily routines.

5. Days of the week, months, and seasons in German.
6. Building simple sentences using the conjugated verb forms and personal pronouns.
7. Grammar practice: Definite and indefinite articles, plural forms.
8. Introducing the verbs “haben” (to have) and “sein” (to be) with conjugation practice.

Activities:

1. **Role-play:** Students practice introducing themselves, asking and answering W-questions.
2. **Group discussion:** Students talk about their hobbies, days of the week, and favorite months/seasons using the vocabulary they learned.
3. **Grammar Quiz:** Personal pronouns, articles, and verb conjugations.

Module-II	**City Life, Directions, and Food**	**6 Hrs.**

Themes:

1. In the city (naming places, buildings, means of transport, basic directions)
2. Food, drink, family, groceries, meals

Grammar:

1. Articles and plural forms
2. Negation (kein, nicht)
3. Imperative forms

Module Content:

1. Vocabulary related to city life: buildings, streets, means of transport.
2. Giving and asking for directions.
3. Learning the imperative mood for giving directions and requests.
4. Vocabulary related to food, meals, and drinks.
5. Talking about family and daily meal routines.
6. Grammar: Using “kein” and “nicht” to form negations.
7. Practice with the accusative case.

Activities:

1. **City tour role-play:** Students practice asking for and giving directions.
2. **Group activity:** Create a menu with German food items, then role-play ordering food.
3. **Grammar exercise:** Negation using "kein" and "nicht."

Module-III	**Everyday Life, Time, Professions, and Health**	**7 Hrs.**

Themes:

1. Everyday life, telling time, making appointments
2. Professions
3. Health and the body

Grammar:

1. Prepositions: “am,” “um,” “von…bis”
2. Modal verbs
3. Possessive articles
4. Perfect tense

Module Content:

1. Telling time and scheduling appointments.

2. Using prepositions (am, um, von…bis) in sentences.
3. Practice with modal verbs for expressing necessity or ability.
4. Talking about professions and workplace vocabulary.
5. Discussing health, body parts, and feelings.
6. Practice using the perfect tense for past actions.

Activities:

1. **Time-based role-play:** Scheduling appointments and practicing telling time.
2. **Profession Bingo:** Students match professions with corresponding vocabulary.
3. **Health questionnaire:** Ask classmates about their health using body-related vocabulary and modal verbs.

Module-IV	**Leisure, Travel**	**6 Hrs.**

Themes:

1. Leisure activities and celebrations
2. Travel, holiday plans, weather

Grammar:

1. Separable verbs
2. Accusative case (continued)
3. Imperative and modal verbs (review)

Module Content:

1. Discussing hobbies, leisure activities, and holiday celebrations.
2. Using separable verbs in the context of free time.
3. Grammar review: Imperative mood, modal verbs.
4. Talking about holiday plans, travel vocabulary, and discussing weather.
5. Review of key grammar concepts throughout the course.

Activities:

Group activity: Plan a holiday trip in German, using travel-related vocabulary and separable verbs.
Weather forecast role-play: Students practice talking about the weather and making holiday plans.
Final review quiz: Comprehensive review of grammar topics such as accusative, modal verbs, perfect tense, and imperative.

Reference Books:

R1: Goyal, M. *Netzwerk: Deutsch als Fremdsprache A1*. Goyal Publishers, 2015.

R2: Schulz-Griesbach: Deutsch als Fremdsprache. Grundstufe in einem Band

Relevant Online Courses (Course name and Weblink)

- NPTEL Course: German - I By Prof. Milind Brahme, IIT Madras, NPTEL
  Link: https://onlinecourses.nptel.ac.in/noc21_hs30/preview
- PICT - Powerlingo Foreign Languages Institute
  Link: https://pict.edu/pict/
- **FACTS ABOUT GERMANY:**
  Link: https://www.tatsachen-ueber-deutschland.de/en
- **ONLINE GERMAN-ENGLISH DICTIONARY:**
  Link: http://www.leo.org/

Second Year B. Tech (S. Y B. Tech) AY (2026-27)

Common to all Programs

[2603OE52]: Foreign Language Studies - Japanese (FLSJ)

Semester	Credits	Teaching Scheme	Examination Scheme
3	2	L: 2 Hrs./ Week	CIE(TW): 50 Marks

Prerequisite: Nil

Course Objectives: The objective of this course is to provide students with

- Enable students to communicate in basic Japanese about themselves and everyday topics.
- Develop an understanding of fundamental Japanese grammar, including particles and basic verb forms.
- Build a vocabulary related to daily life, city environments, food, leisure, and travel.
- Introduce students to aspects of Japanese culture and customs.

Course Outcomes: After completing this course, students will be able to

CO1: Introduce themselves and others, and talk about their hobbies in Japanese.
CO2: Describe places in the city, give directions, and order food in Japanese.
CO3: Discuss daily routines, professions, and basic health in Japanese.
CO4: Talk about their leisure activities and travel plans in Japanese.

COURSE CONTENTS

Module-I	Introduction, Personal Information, and Basic Grammar	7 Hrs.

Themes:

1. Introduction to Japanese scripts (Hiragana, Katakana)
2. Introducing oneself and others (name, nationality, etc.)
3. Hobbies

Grammar:

1. Basic sentence structure (Subject-Object-Verb)
2. Particles: wa (は), ga (が), mo (も)
3. Pronouns: watashi (私), anata (あなた)
4. Counters (basic introduction)

Module Content:

1. Introduction to Hiragana and Katakana, basic stroke order and pronunciation.
2. Greetings and introductions: Hajimemashite, Yoroshiku onegaishimasu.
3. Using particles to indicate the topic and subject of a sentence.
4. Talking about hobbies using simple sentence structures.
5. Counting simple objects (using basic counters).

Activities:

1. **Writing practice:** Hiragana and Katakana characters.
2. **Role-play:** Introducing oneself to a classmate and asking about hobbies.
3. Counting objects in the classroom (e.g., pencils, books).

Module-II	City Life, Directions, and Food	6 Hrs.

Themes:

1. Places in the city (train station, school, supermarket, etc.)
2. Asking for and giving directions
3. Food and drinks

Grammar:

1. Locational particles: ni (に), e (へ)
2. Directional words: migi (右), hidari (左), mae (前), ushiro (後ろ)
3. Verb arimasu/imasu (あります/います)

Module Content:

1. Vocabulary for common places in a city.
2. Giving and understanding basic directions using landmarks.
3. Talking about food and drinks, ordering in a restaurant.
4. Using arimasu/imasu to indicate the existence of things/people.

Activities:

1. **City map activity:** Pointing out places and giving directions.
2. **Restaurant role-play:** Ordering food and drinks.
3. Describing the contents of a room using arimasu/imasu.

Module-III	Everyday Life, Time, Professions, and Health	7 Hrs.

Themes:

1. Daily routines
2. Telling time and making appointments
3. Professions
4. Basic health vocabulary

Grammar:

1. Time expressions: ji (時), fun (分), gozen (午前), gogo (午後)
2. Verb conjugation (present and past tense)
3. Particles kara (から) and made (まで) to indicate time duration

Module Content:

1. Describing daily routines using time expressions and verbs.
2. Asking about and stating professions.
3. Basic vocabulary related to health and common ailments.
4. Making simple appointments.

Activities:

1. **Daily routine presentation:** Describing one's daily schedule.
2. **Role-play:** Making an appointment with a doctor.
3. **Profession guessing game.**

Module-IV	Leisure, Travel	6 Hrs.

Themes:

1. Hobbies and leisure activities

2. Travel and holiday plans
3. Weather

Grammar:

1. ~tai desu (〜たいです) to express desires
2. Adjectives (review and expansion)
3. Conditional form ~tara (〜たら) for hypothetical situations

Module Content:

1. Talking about hobbies and things you want to do.
2. Describing travel plans and destinations.
3. Talking about the weather.
4. Using conditional sentences to express hypothetical travel scenarios.

Activities:

1. **Holiday plan presentation:** Describing a dream vacation.
2. **Role Play:** Weather forecast.
3. **Sentence construction:** Expressing desires and hypothetical situations using ~tai desu and ~tara.

Reference Books:

R1: Yamamoto, N. *Shin Nihongo no Kiso I (Romanized Edition)*. Association for Overseas Technical Scholars (AOTS), 3A Corporation, June 1990.

R2: *Minna no Nihongo*. 3A Network, Goyal Publishers.

R3: Mizutani, Osamu, and Nobuko Mizutani. *Introduction to Modern Japanese*. Japan Times, November 1992.

R4: Nichimo, A. *250 Essential Kanji for Everyday Use*. 2nd rev. ed., Tuttle Publishing, January 2004.

R5: *Japanese for Busy People*. 3rd ed., Association for Japanese Language Teaching, Kodansha Tokyo, Kodansha International, November 2011.

Relevant Online Courses (Course name and Weblink)

- NPTEL Course: Introduction to Japanese Language and Culture by Prof. Vatsala Misra, IIT Kanpur Link: https://onlinecourses.nptel.ac.in/noc19_hs52/preview
- PICT - Powerlingo Foreign Languages Institute
  Link: https://pict.edu/pict/

Second Year B-Tech (SY B. Tech) AY (2026-27)
Common to all Programs

[2603VE11]: Universal Human Values (UHV)

Semester	Credits	Teaching Scheme	Examination Scheme
3	2	**L: 02 Hrs./ Week**	**CIE (TW): 25 Mark**

Prerequisite: UHV-I: Universal Human Values-Introduction (SIP)

Course Objectives:

The objective of this course is to provide students with:

- An appreciation for the essential complementarity between 'values' and 'skills' as a foundation for sustained happiness and prosperity — the core aspirations of every human being
- To facilitate the development of a Holistic perspective among students towards life and profession as well as towards happiness and prosperity based on a correct understanding of the Human reality and the rest of existence.
- Insights into the practical implications of a holistic understanding — fostering ethical human conduct, trustful and fulfilling relationships, and mutually enriching interactions with nature.

Course Outcomes: **After completion of this course, students will be able to**

CO1: Distinguish among values and skills; happiness and physical facilities; the Self and the Body; and understand the importance of intention, competence, trust, and respect in building harmonious personal and professional relationships.

CO2: Analyze the role of human beings in ensuring harmony in relationships, society, and nature, and apply ethical principles with trust and respect for sustainable living and professional conduct.

COURSE CONTENT

Module-I	**Basic aspiration of Human being & Harmony in Human being**	**12 Hrs**

Understanding Value Education, Self-exploration as the Process for Value Education, Continuous Happiness and Prosperity – the Basic Human Aspirations, Right Understanding, Relationship and Physical Facility, Happiness and Prosperity – Current Scenario, Method to fulfill the Basic Human Aspirations. Understanding Human being as the Co-existence of the Self and the Body, distinguishing between the Needs of the Self and the Body, The Body as an Instrument of the Self, Understanding Harmony in the Self, Harmony of the Self with the Body, Program to ensure self-regulation and Health.

Module-II	**Harmony in the Family, society & Nature / Existence**	**12 Hrs**

Harmony in the Family – the Basic Unit of Human Interaction, Values in Human–to–Human Relationship, Nine universal values in relationships viz. Trust, Respect, Affection, Care, Guidance, Reverence, Glory, Gratitude, Love. Understanding Harmony in Society, Vision for the Universal Human Order, Human Order Five Dimension. Understanding Harmony in Nature, self–regulation & mutual fulfillment among the Four orders of Nature, Realizing Existence as co-existence at all levels holistic perception of harmony in existence.

Textbooks:

T1.	Gaur, R. R., Sangal, R., and Bagaria, G. P. Human Values and Professional Ethics 3rd revised ed., PHI, Excel Books Pvt. Ltd., New Delhi, 2010.
Reference Books:	
R1.	A Nagaraj, Jeevan Vidya: Ek Parichaya, Jeevan Vidya Prakashan, Amarkantak, 1999.
R2.	A. N. Tripathi, Human Values, New Age Intl. Publishers, New Delhi, 2004
R3.	Mohandas Karamchand Gandhi, “The Story of My Experiments with Truth”, Fingerprint Publishing, ISBN: 978-8172343118.
R4.	Dharampal “Education Rediscovering India”, Stosius Inc/Advent Books Division Publishing, ISBN: 978-0706922776
R5.	Mohandas K. Gandhi, “Hind Swaraj or Indian Home Rule”, ISBN :978-1449922214
Websites and Online Resources:	
W1.	**Universal Human Values** ● Website: Universal Human Values - YouTube
W2.	**English eSIP Module 1 Universal Human Values I(UHV I) Session 1& 2** ● Website: https://www.youtube.com/live/OgdNx0X923I?feature=shared
Relevant MOOCs Course (Course name and Weblink)	
● Swayam Course on “Understanding Human Being Nature and Existence Comprehensively” by Dr. Kumar Sambhav, Director, UP Institute of Design (UPID), Noida. **Link:** Understanding Human Being Nature and Existence Comprehensively - Course ● NPTEL Course on “Exploring Human Values: Visions of Happiness and Perfect Society” by Prof. A. K. Sharma, Department of Humanities and Social Sciences, IIT Kanpur. **Link:** nptel.ac.in/courses/109104068	
Relevant Topic for Self-study:	
1. Making the Right Choices: Staying True to Your Values Despite Outside Pressure 2. How Kindness and Understanding Help Build Strong Relationships	

Second Year B. Tech (S. Y B. Tech) AY (2026-27)
Common to all Programs

[2603AE31]: Professional Development and Career Readiness (PDCR)

Semester	Credits	Teaching Scheme	Examination Scheme
3	**1**	**P: 2 Hrs./ Week**	**CIE (TW): 25 Marks**

Prerequisite: Students should have prior knowledge of

1. Soft Skills (SS) from FY B. Tech Courses

Course Objectives: The objective of this course is to provide students with

- The skills to prepare a good resume, as well as prepare for interviews and group discussions.
- The ability to explore desired career opportunities in the employment market while considering their personal strengths, weaknesses, opportunities, and challenges (SWOC).
- The necessary career skills to partake in and fully pursue a successful career path.

Course Outcomes: After completing this course, students will be able to

CO1: Prepare the resume on an appropriate template without any grammatical and syntax errors, and Present and Discuss with students.
CO2: Participate in a simulated interview and evaluate your own performance for betterment.
CO3: Demonstrate effective communication skills through Group Discussion, self-management attributes.
CO4: Define personal and career goals (short-term and long-term) using introspective skills and Perform SWOC assessment.
CO5: Identify career opportunities in consideration of potential and aspirations.

COURSE CONTENTS

Expt. No.	Task to carry out	Hrs.	CO
1.	**Resume Skills** ● Introduction of resume and its importance ● Difference between a CV, resume and biodata ● Essential components of a good resume. ● Common errors while preparing a resume		CO1
2.	Prepare a good resume considering all essential components and present the resume		CO 1
3.	**Interview Skills: Preparation and Presentation** ● Meaning and types of interviews (F2F, telephonic, video, etc.) ● Dress code, background research, dos and don'ts. ● Situation, task, action, and response (STAR concept) for facing an interview. ● Interview procedure (opening, listening skills, and closure). ● Important questions generally asked at a job interview (open- and close-ended questions)		CO 2
4.	**Interview Skills: Common Errors** ● Discuss the common errors that candidates generally make at an interview ● Demonstrate an ideal interview		CO 3

5.	**Group Discussion Skills** ● Meaning and Methods of Group Discussion ● Procedure of Group Discussion ● Group Discussion — Simulation ● Group Discussion — Common Errors	CO 3
6.	**Strengths, Weaknesses, Opportunities and Challenges Analysis (SWOC):** ● To carry out introspection and become aware of one’s Strengths, Weakness, ● Opportunities and Threats. ● Document SWOT analysis in a matrix format.	CO 4
7.	**Exploring Career Opportunities** ● Knowledge about the world of work, requirements of jobs, including self-employment. ● Sources of career information. ● Preparing for a career based on potential and availability of opportunities.	CO 5

Text Books:

T1.	Bhattacharya, I. *An Approach to Communication Skills*. Dhanpat Rai.
T2.	Chauhan, R. G. S., and Sharma, S. *Soft Skills: An Integrated Approach to Maximize Personality*. Wiley, First Edition, 2016.

Reference Books:

R1.	Sweeney, S. *English for Business Communication*. Cambridge University Press.
R2.	Kumar, S., and Lata, P. *Communication Skills*. Oxford University Press.
R3.	Kalam, A. P. J. *Ignited Minds: Unleashing the Power Within India*. Penguin Books India, New Delhi, 2003.

Relevant Topics for Self-study:

- Foundation Skills in IT (FSIT) — Refer to the websites like https://www.sscnasscom.com/ssc-projects/capacity-building-and-development/training/fsit/ and
- Global Business Foundation Skills (GBFS) – Refer websites like https://www.sscnasscom.com/ssc-projects/capacity-building-and-development/training/gbfs/

Second Year B. Tech (S. Y B. Tech) AY (2026-27)
Common to all Programs

[2603CE31]: Community Engagement Project (CEP)

Semester	Credits	Teaching Scheme	Examination Scheme
3	1	P: 2 Hrs./ Week	CIE (TW): 25 Marks

Prerequisite: Students should have prior knowledge of

- Basic understanding of social and ethical responsibilities.
- Teamwork and communication skills acquired in prior coursework or group activities.
- Familiarity with problem-solving methodologies and project planning.

Course Objectives: The objective of this course is to provide students with

- Opportunities to engage with their local community, fostering empathy, teamwork, and problem-solving skills while contributing positively to their surroundings.
- An understanding of the challenges faced by the local community and the role of engineering in addressing those challenges.
- The ability to apply technical knowledge and skills to design solutions or interventions that create a positive impact on the community.
- The skills to evaluate and critically analyze the outcomes of their engagement activities, deriving actionable insights for sustainable impact.

Course Outcomes: After completing this course, students will be able to

CO1: Identify and Analyze community needs and challenges by engaging with stakeholders and evaluating real-world problems. *(Remembering & analyzing)*

CO2: Design and Implement practical, creative, and context-specific solutions using engineering principles to address community issues. *(Creating & applying)*

CO3: Reflect and Evaluate the effectiveness of their interventions and articulate lessons learned through reports and presentations. *(Evaluating & Understanding)*

COURSE GUIDELINES

A community engagement project is intended to instill social responsibility and to connect students with local communities to address real-life challenges and promote sustainable development. Students are expected to contribute to the community by sharing their learning outcomes and solve/propose solutions to societal/community problems. The motto of the community engagement project is 'Campus to Community'. Students are expected to identify socially relevant problems/projects under the guidance of teachers and solve or propose solutions. These projects foster collaboration, empathy, and social responsibility.

Projects may include, but not limited to, diverse areas such as health, where students can organize free check-up camps or mental health awareness drives; livelihood, through skill-sharing or micro-entrepreneurship support; and education, via digital literacy workshops, mobile libraries, or career guidance camps. Environmentally impactful projects include rainwater harvesting awareness and solar lighting in villages. Moreover, projects like documenting local history or organizing cultural exchange events help preserve and celebrate community identity. Such initiatives not only benefit society but also provide participants with practical experience, leadership skills, and a deeper understanding of civic duties. Through these engagements, communities become active partners in development, creating a more inclusive and resilient society.

A. Group Formation:

- Form a group of 3-4 students that share a similar interest in each batch, Duration: 24 hours (divided into manageable sessions or shifts).
- The group should be cohesive, sharing and caring, and contribute to the task assigned.
- The tasks carried out need to be maintained in the LOG book by each group.

B. Project Scope:

The CEP should focus on addressing a specific community or societal issue. Projects may fall under the following themes:

1. Education and Awareness:

- Conduct workshops or awareness drives on topics like digital literacy, environmental sustainability, mental health, or career planning for local stakeholders.

2. Technology for Social Good:

- Develop a simple prototype or solution that addresses a real-world problem (e.g., a water-saving device, simple mobile apps, or tools for community use).

3. Environmental Sustainability:

- Organize clean-up drives, tree plantations, recycling campaigns, or energy conservation initiatives.

4. Health and Wellness:

- Promote health through awareness programs on hygiene, nutrition, and exercise.

5. Skill Development:

- Teach basic computer or technical skills to students, staff, or the community.

C. Step-by-Step Execution Plan:

1. Planning Phase:

- **Team Formation:**

Form teams of 3-4 students with a balance of skills and interests.

- **Project Selection:**

Choose a project theme and define a clear objective that aligns with community needs.

- **Proposal Submission:**
  - Submit a one-page project proposal outlining:
  - Title of the project.
  - Objective and expected outcome.
  - Plan of execution (timeline and activities).
  - Required resources (if any).
  - Get approval from the designated faculty mentor.

2. Execution Phase:

- **Phase 1 Activities**

- Conduct initial outreach and engage with the community or target participants.
- Implement planned activities with close teamwork and documentation.

- **Phase Activities**
  - Continue engagement and collect feedback from the participants.
  - Begin summarizing the outcomes of the project.
- **Best Practices:**
  - Maintain a positive attitude and open communication with the community.
  - Respect cultural norms and values of the participants.
  - Adapt your plan based on real-time needs or challenges.

3. Reporting Phase:

- **Documentation:**
  - Create a detailed report containing
  - Title, objective, and scope of the project.
  - Activities conducted and timeline.
  - Outcomes and community feedback.
  - Photos/videos of the activities (if permitted).
  - Challenges faced and how they were addressed.
- **Presentation:**
  - Each team will present their project to a panel of faculty members or peers, showcasing their efforts and outcomes.
  - Duration of presentation: 5-7 minutes per team.

D. Evaluation Criteria:

Projects will be evaluated based on:

1. **Relevance:** How well the project aligns with community needs.
2. **Impact:** The tangible and intangible benefits delivered to the community.
3. **Innovation:** Creativity in the approach or solution provided.
4. **Teamwork:** Collaboration and effective delegation within the group.
5. **Documentation & Presentation:** Clarity, depth, and overall delivery of the report and presentation.

E. Guidelines for Conduct:

1. **Behavior:** Students should display professionalism, punctuality, and respect.
2. **Safety:** Follow all safety protocols during on-campus or fieldwork activities.
3. **Feedback:** Collect feedback from participants to measure the success and identify areas for improvement.

F. Support and Supervision:

1. Faculty mentors will be assigned to each group to guide them throughout the project.

2. A resource or helpdesk will be available for logistical or technical support.
Reference Books:
R1. Dostilio, L. D., et al. *The Community Engagement Professional's Guidebook: A Companion to The Community Engagement Professional in Higher Education*. Stylus Publishing, 2017. A practical guide for community engagement projects, including tools and strategies for effective implementation and assessment.
R2. Waterman, A. *Service-Learning: A Guide to Planning, Implementing, and Assessing Student Projects*. Routledge, 1997. Insights into service-learning methodology, planning, and assessment techniques for impactful projects.
R3. Beckman, M., and Long, J. F. *Community-Based Research: Teaching for Community Impact*. Stylus Publishing, 2016. Approaches for conducting research and engagement projects collaboratively with communities.
R4. IDEO.org. *Design Thinking for Social Innovation*. IDEO Press, 2015. Explains how to apply design thinking to solve social problems, ideal for projects focusing on community engagement.
R5. Sherrod, L. R., Torney-Purta, J., and Flanagan, C. A. (Eds.). *Handbook of Research on Civic Engagement in Youth*. Wiley, 2010. A detailed guide on youth involvement in civic and community projects, with case studies and strategies for engagement.
Websites and Online Resources:
For Planning and Conducting Projects:
W1. UNESCO: Education for Sustainable Development ● Website: https://www.unesco.org ● Focus: Resources and case studies related to sustainability and community engagement.
W2. EPICS (Engineering Projects in Community Service) ● Website: https://engineering.purdue.edu/EPICS ● Focus: Offers methodologies and tools for engineering students to work on real-world projects benefiting communities.
W3. Ashoka: Innovators for the Public ● Website: https://www.ashoka.org ● Focus: Information on social entrepreneurship and community innovation projects.
W4. Design for Change ● Website: https://www.dfcworld.com ● Focus: Templates, toolkits, and project ideas for implementing impactful community-based projects.
For Evaluation and Impact Assessment:
W5. Community Tool Box (University of Kansas) ● Website: https://ctb.ku.edu ● Focus: Comprehensive resources for community engagement, project evaluation, and measuring outcomes.
W6. UN SDG (Sustainable Development Goals) Knowledge Platform ● Website: https://sdgs.un.org/ ● Focus: Guidance on aligning community engagement projects with UN Sustainable Development Goals (SDGs).
W7. Campus Compact

- **Website: https://www.compact.org/**
- **Focus: Resources on civic and community engagement for students and educators, with a focus on project assessment.**

W8. BetterEvaluation

- Website: https://www.betterevaluation.org
- Focus: Tools and frameworks to evaluate the impact of community projects effectively.

W9. Plan-Do-Check-Act Cycle (PDCA) – Deming Institute

- Website: https://deming.org/explore/pdsa
- Focus: Step-by-step guides for planning, implementing, and refining community projects.

Relevant MOOCs Course (Course name and Weblink)

1. NPTEL course: Ecology and Society, by Prof. Ngamjahao Kipgen, IIT Guwahati
   This course delves into the dynamic relationships between human cultures and their ecological environments, focusing on human-environment interactions and sustainable development.
   Link: https://onlinecourses.nptel.ac.in/noc20_hs77/preview.
2. NPTEL course: Basics of Health Promotion and Education Intervention, by Dr. Arista Lahiri, Dr. Sweety Suman Jha (IIT Kharagpur), Dr. Madhumita Dobe, Dr. Chandrashekhar Taklikar (AIIH&PH, Kolkata)
   This course provides a comprehensive understanding of health promotion and education interventions, covering planning, implementation, and evaluation strategies.
   Link: https://onlinecourses.nptel.ac.in/noc22_ge18/preview
3. NPTEL course: A Hybrid Course on Water Quality – An Approach to People's Water Data, by IIT Madras
   This hybrid course emphasizes practical fieldwork, including water sample collection and analysis, engaging with communities to assess water quality.
   Link:
   https://elearn.nptel.ac.in/shop/iit-workshops/completed/a-hybrid-course-on-water-quality-an-approach-to-peoples-water-data/?v=c86ee0d9d7ed

Second Year B-Tech
(S. Y B. Tech)

Semester-4

Second Year B. Tech (S. Y B. Tech) AY (2026-27)
Common to all Programs

[2604AE32]: Collaborative Skills, Digital Ethics, and Cyber Security (CDC)

Semester	Credits	Teaching Scheme	Examination Scheme
4	1	P: 2 Hrs./ Week	CIE (TW): 25 Marks

Prerequisite: Students should have prior knowledge of

- Course on Soft Skills (SS)

Course Objectives: The objective of this course is to provide students with

- Recognize the importance of team skills and develop strategies to acquire them.
- Effectively design, develop, and adapt to various situations both individually and as part of a team.

Course Outcomes: After completing this course, students will be able to

CO1: Empathize with and trust colleagues for improving interpersonal relations.

CO2: Demonstrate effective communication by respecting diversity and embracing good listening skills.

CO3: Distinguish the guiding principles for communication in a diverse, smaller, internal world.

CO4: Carry out ethical practices for interpersonal skills for better social and professional relations with seniors, juniors, peers, and stakeholders. Also, inculcate digital ethics and cyber security measures and practices.

COURSE CONTENTS

Expt. No.	Task to carry out	CO
1.	**Trust and Collaboration** Explain the Importance of Trust in Creating a Collaborative Team Agree to Disagree and Disagree to Agree – Spirit of Teamwork Understanding Fear of Being Judged and Strategies to Overcome Fear.	CO1
2.	**Listening as a Team Skill** Advantages of Effective Listening Listening as a Team Member and Team Leader. Use of active listening strategies to encourage sharing of ideas (full and undivided attention, no interruptions, no pre-think, use empathy, listen to tone and voice modulation, recapitulate points.).	CO 2
3.	**Brainstorming** Brainstorming as a Technique to Promote Idea Generation a. Brainstorming: Meaning and the Process b. Procedure for Conducting Brainstorming c. Importance of Using Brainstorming Technique d. Types of Brainstorming	CO 3
4.	Learning and Showcasing the Principles of Documentation of Team Session Outcomes.	CO 3
5.	**Social and Cultural Etiquette** Need for Etiquette (impression, image, earn respect, appreciation) • Aspects of Social and Cultural/Corporate Etiquette in Promoting Teamwork • Importance of Time, Place, Propriety and Adaptability to Diverse Cultures	CO 4
6.	**Digital Ethics**	CO 4

	Digital Ethics i. Digital Literacy Skills, ii. Digital Etiquette, iii. Digital Life Skills	
7.	**Cyber Security** The Art of Protecting Secrets a. Understanding Encryption and Decryption and Its Different Types b. Art of Data Masking c. Firewall and Its Proper Use in Cyber Protection	CO 4

Text Books:

T1.	Ratliff, J., *Leadership Through Trust & Collaboration: Practical Tools for Today's Results-Driven Leader*, Morgan James Publishing, 2020.
T2.	Dauda, J., *Cybersecurity and Digital Ethics: Principles of Cybersecurity (Cybersecurity Practices, Technologies, and Processes), 2023*.

Reference Books:

R1.	Kelly, T., & Kelly, D., *Creative Confidence: Unleashing the Creative Potential Within Us All*, Harper Collins Publishers India, New Delhi, 2014.
R2.	Sweeney, S., *English for Business Communication*, Cambridge University Press, 2003.
R3.	Kumar, S., & Lata, P., *Communication Skills*, Oxford University Press, 2015.

Students can avail additional resources to enhance soft skills further

1. **Leadership**, by Prof. Kalyan Chakravarti and Prof. Tuheena Mukherjee, IIT Kharagpur
   Link: https://onlinecourses.nptel.ac.in/noc19_mg34/preview.
2. **Towards an Ethical Digital Society: From Theory to Practice**, by Prof. Bidisha Chaudhuri, IIIT Bangalore
   Link: https://nptel.ac.in/course s/109106184
3. **Global Business Foundation Skills (GBFS)** – Refer websites like https://www.sscnasscom.com/ssc-projects/capacity-building-and-development/training/gbfs/

Second Year B. Tech (S. Y B. Tech) AY (2026-27)
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[2604VE12]: Indian Constitution and Social Responsibility (ICSR)

Semester	Credits	Teaching Scheme	Examination Scheme
4	1	L: 1 Hrs./ Week	CIE (TW): 25 Marks

Prerequisite: Students should have prior knowledge of

- Basic Knowledge of Civics and Governance.
- Ethical Reasoning and Social Awareness, Communication and Critical Thinking Skills.

Course Objectives: The objective of this course is to provide students with

- An understanding of the principles of social responsibility, ethical citizenship, and the Indian Constitution.
- The ability to analyze the role of individuals and institutions in fostering responsible citizenship, democracy, and social change.
- Skills to evaluate ethical dilemmas and legal frameworks for making informed civic decisions.
- Opportunities to design initiatives that promote social responsibility and active community participation.

Course Outcomes: After completing this course, students will be able to

CO1: Explain fundamental concepts of social responsibility, civic engagement, and constitutional law.

CO2: Apply ethical and legal principles to address community and global issues.

CO3: Analyze the relationship between fundamental rights, duties, and governance in India.

CO4: Propose community-driven ideas that contribute to sustainable development and civic well-being.

COURSE CONTENTS

Module-I	Introduction to Indian Constitution	4 Hrs.

1. Historical Background and Evolution of the Indian Constitution
2. Preamble and its significance
3. Fundamental Rights and Duties
4. Directive Principles of State Policy

Activities:

1. **Debate:** Relevance of Fundamental Rights in Contemporary India
2. **Case Study:** Landmark Supreme Court Judgments

Module-II	Government Structure & Electoral System	4 Hrs.

1. Separation of Powers: Legislature, Executive, and Judiciary
2. Parliamentary vs. Presidential System
3. Supreme Court and High Court
4. Federalism: Centre-State Relations
5. Election Commission and Electoral Reforms (Antidefection law)

Activities:

1. **Mock Parliament Session**
2. **Discussion:** Impact of Electoral Reforms on Indian Democracy. Role of executives.

Module-III	**Social Responsibility & Citizenship**	**3 Hrs.**

1. Definitions of Social Responsibility and Citizenship
2. Ethics and Moral Duties in Society
3. Individual vs. Collective Responsibility
4. Case Studies: Impactful Citizens and Social Movements

Activities:

1. **Group Discussion:** What does responsible citizenship mean to you?
2. **Reflection Assignment:** Personal Social Responsibility

Module-IV	**Civic Engagement & Sustainable Development**	**3 Hrs.**

1. Forms of Civic Engagement (Volunteering, Advocacy, Social Activism)
2. Role of NGOs, Government, and Private Sectors
3. Sustainable Development Goals (SDGs)
4. Corporate Social Responsibility (CSR)

Activities:

1. **Role-Playing Exercise:** Simulating a Town Hall Meeting
2. **Local Community Service Initiative**

Reference Books:

R1:	Sen, Amartya. *The Idea of Justice,* Discusses fairness and ethics in society, 2009.
R2:	D.D. Basu, *Introduction to the Constitution of India*, LexisNexis, Latest Edition.
R3:	Granville Austin, *The Indian Constitution: Cornerstone of a Nation*, Oxford University Press.
R4:	Rawls, John. *A Theory of Justice*– Covers principles of justice and democracy, 1971.
R5:	United Nations Sustainable Development Goals (SDGs) – Official UN resources on social responsibility.
R6:	Sachs, Jeffrey. *The Age of Sustainable Development*– Insights into global responsibility, 2015.

Relevant Online Courses (Course name and Weblink)

- **Harvard University (edX): "Justice" by Michael Sandel** – Ethics & civic responsibility.
- **Coursera (University of London): "Global Diplomacy – The United Nations in the World"** – Understanding international citizenship.
- **Future Learn: "Social Responsibility and Sustainable Development"** – Corporate & personal social responsibility.
- **Khan Academy: "Civics & Government"** – Basic concepts of democracy and civic engagement.

Relevant Topics for Self-study:

- **Corporate Social Responsibility,** by Prof. Aradhna Malik, IIT Kharagpur
  This course introduces participants to the field of Corporate Social Responsibility (CSR), covering its history, planning, implementation, evaluation, and future directions.
  Link: Corporate Social Responsibility
- NPTEL course: **Community Engagement and Social Responsibility,** by Prof. Akshay Kumar Satsangi, Dayalbagh Educational Institute, Agra
  This course emphasizes the importance of community development through self-help groups, health and well-being, literacy, employment, and the role of social networking in bridging government schemes and the people of India.

Link: Community Engagement and Social Responsibility. ● NPTEL course: **Constitutional Government & Democracy in India,** by Prof. Amitabha Ray, St. Xavier's College (Autonomous), Kolkata This course acquaints students with the constitutional design of state structures and institutions, and their actual working overtime. It traces the embodiment of conflicting impulses within the constitution and encourages a study of state institutions in their mutual interaction and with the larger extra-constitutional environment. **Link:** SWAYAM: Constitutional Government & Democracy in India ● NPTEL course: **Constitution Law and Public Administration in India,** By Prof. Sairam Bhat, National Law School of India University This course explores the intricacies of constitutional law and public administration in India, providing insights into the legal frameworks and administrative structures that govern the country. **Link:** NPTEL: Constitution Law and Public Administration in India
Any special topics of interest: Constitutional Bodies, Competitive examinations: UPSC, MPSC, IES.